

Performance Analysis

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Detailed analysis of your performance in the AWS Certified Solutions Architect Associate - SAA C03 exam.

Answer Distribution

0

Correct

0

Incorrect

65

Unattempted

Domain Performance

Automation	0%
Security	0%
Databases	0%
Networking	0%
Storage	0%
Compute	0%
Serverless	0%



Final Score

**0/65**

Passing score: 48

Time Taken

**130m 0s**

Total time: 130 minutes

Question Analysis

Correct	0
Incorrect	0
Unattempted	65
Reviewed	0

Question-wise Analysis

Expand All

Collapse All

Question 1

Not Attempted

Automation



An ecommerce company needs to build and train machine learning (ML) models to visualize complex scenarios and detect trends in customer data. The architecture team wants to integrate these models with a reporting platform for analysis and direct use in business intelligence dashboards. Which solution meets these requirements with the LEAST operational overhead?

- ☐ Use AWS Glue to create an ML transform to build and train models. Use Amazon OpenSearch Service to visualize the data.
AWS Glue's ML Transforms are primarily for data cleaning and deduplication, not for building complex models. OpenSearch is not optimized for business intelligence visualizations compared to QuickSight.
- ☒ Use Amazon SageMaker to build and train models. Use Amazon QuickSight to visualize the data.
Amazon SageMaker provides a fully managed environment for building, training, and deploying ML models. Amazon QuickSight is a cloud-native BI service ideal for creating interactive visualizations and dashboards.
- ☐ Use a pre-built ML Amazon Machine Image (AMI) from the AWS Marketplace to build and train models. Use Amazon OpenSearch Service to visualize the data.
Using a pre-built AMI adds operational overhead for managing and patching the underlying infrastructure. OpenSearch is not optimized for business intelligence visualizations compared to QuickSight.
- ☐ Use Amazon QuickSight to build and train models by using calculated fields. Use Amazon QuickSight to visualize the data.
Amazon QuickSight's calculated fields are for data transformation and analysis, not for building and training complex ML models.

Overall Explanation:

Amazon SageMaker is a fully managed service, simplifying the ML workflow. Amazon QuickSight integrates seamlessly with SageMaker and is purpose-built for business intelligence visualizations and dashboards. This combination offers the least operational overhead for the given requirements.

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Question 2

Not Attempted

Security



A company uses multiple AWS accounts organized within AWS Organizations for its production and non-production environments. They need to prevent unauthorized modification of cost allocation tags across all accounts in the organization, while still allowing authorized principals

to modify them. Which solution provides the MOST effective and scalable way to meet these requirements?

- ⊗ Create a custom AWS Config rule to detect and remediate tag modifications except by authorized principals.

AWS Config can detect configuration changes, but it's not the ideal solution for preventing modifications across an entire AWS Organization. It's better suited for enforcing specific configurations within an account, rather than implementing organization-wide guardrails. Remediation would also be reactive, not preventative.

- ⊗ Create a custom trail in AWS CloudTrail to log tag modification events and trigger alerts.

CloudTrail provides detailed logging of API calls, including tag modifications. However, it's a monitoring tool and doesn't prevent modifications. While alerts can be triggered, they are reactive and don't enforce preventative controls.

- ✔ Create a service control policy (SCP) to prevent tag modification except by authorized principals.

Service Control Policies (SCPs) are the correct solution. SCPs allow you to define AWS permissions guardrails at the AWS Organizations level, affecting all accounts within the organization. You can create an SCP that denies tag modification actions for all principals except those you explicitly authorize, providing a centralized and preventative control.

- ⊗ Create custom Amazon CloudWatch logs and alarms to detect tag modification attempts.

Amazon CloudWatch is primarily a monitoring and alerting service. While it can ingest logs and trigger alarms based on tag modification events, it doesn't prevent the modifications from occurring in the first place. It's a reactive solution, not a preventative one.

Overall Explanation:

Service Control Policies (SCPs) are the most effective solution to prevent tag modification across multiple AWS accounts within an AWS Organization. SCPs define guardrails at the organization level, restricting the actions that IAM users and roles can perform in member accounts. By creating an SCP that denies tag modification actions except for authorized principals, the company can ensure consistent enforcement of tag policies across all accounts. AWS Config and CloudWatch logs are useful for monitoring and auditing, but do not prevent unauthorized actions. CloudTrail provides logging but does not prevent modifications.

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Question 3

Not Attempted

Automation



A company hosts its application in the AWS Cloud. The application runs on Amazon EC2 instances behind an Elastic Load Balancer in an Auto Scaling group and with an Amazon DynamoDB table. The company wants to ensure the application can be made available in another AWS Region with minimal downtime. What should a solutions architect do to meet these requirements with the LEAST amount of downtime?

- ☒ Create an Auto Scaling group and a load balancer in the disaster recovery Region. Configure the DynamoDB table as a global table. Configure DNS failover to point to the new disaster recovery Region's load balancer.

This option pre-deploys the infrastructure in the DR region, ensuring that it is readily available and minimizing the downtime during a failover. The DNS failover provides the fastest way to redirect traffic.

- ☐ Create an AWS CloudFormation template to create EC2 instances, load balancers, and DynamoDB tables to be launched when needed. Configure DNS failover to point to the new disaster recovery Region's load balancer.

While CloudFormation is useful, creating resources on demand will result in more downtime than having them pre-provisioned.

- ☐ Create an AWS CloudFormation template to create EC2 instances and a load balancer to be launched when needed. Configure the DynamoDB table as a global table. Configure DNS failover to point to the new disaster recovery Region's load balancer.

Similar to Option B, this option also incurs significant downtime as the EC2 instances and load balancer are created only when needed.

- ☐ Create an Auto Scaling group and load balancer in the disaster recovery Region. Configure the DynamoDB table as a global table. Create an Amazon CloudWatch alarm to trigger an AWS Lambda function that updates Amazon Route 53 pointing to the disaster recovery load balancer.

Although it sets up the Auto Scaling group and load balancer for rapid deployment, using CloudWatch and Lambda adds unnecessary complexity and potential delays in DNS propagation compared to Route 53 DNS Failover feature.

Overall Explanation:

Option A provides the least amount of downtime by pre-provisioning the necessary infrastructure (Auto Scaling group, load balancer) in the DR region and utilizing DynamoDB Global Tables for data replication. Route 53 DNS failover offers the quickest method to shift traffic to the DR environment in case of a failure.

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Question 4

Not Attempted

Databases



A company needs to migrate a 20 TB MySQL database from its on-premises data center to AWS within 2 weeks. The company wants to complete the migration with minimal downtime and is looking for the MOST cost-effective solution. Which of the following options should they choose?

- ☒ Order an AWS Snowball Edge Storage Optimized device. Use AWS Database Migration Service (AWS DMS) with AWS Schema Conversion Tool (AWS SCT) to migrate the database with replication of ongoing changes. Send the Snowball Edge device to AWS to finish the migration and continue the ongoing replication.

This is the most cost-effective solution for migrating a large database within the given timeframe and with minimal downtime. Snowball Edge allows for fast data transfer, while DMS and SCT handle schema conversion and ongoing replication.

- ☐ Order an AWS Snowmobile vehicle. Use AWS Database Migration Service (AWS DMS) with AWS Schema Conversion Tool (AWS SCT) to migrate the database with ongoing changes. Send the Snowmobile vehicle back to AWS to finish the migration and continue the ongoing replication.

Snowmobile is designed for petabyte-scale data transfers and is not cost-effective for a 20 TB database.

- ☐ Order an AWS Snowball Edge Compute Optimized with GPU device. Use AWS Database Migration Service (AWS DMS) with AWS Schema Conversion Tool (AWS SCT) to migrate the database with ongoing changes. Send the Snowball device to AWS to finish the migration and continue the ongoing replication.

The Compute Optimized Snowball Edge is not as cost-effective as the Storage Optimized version for this data transfer scenario, as the compute resources are not required.

- ☐ Order a 1 GB dedicated AWS Direct Connect connection to establish a connection with the data center. Use AWS Database Migration Service (AWS DMS) with AWS Schema

Conversion Tool (AWS SCT) to migrate the database with replication of ongoing changes.

Setting up Direct Connect takes longer than 2 weeks, making it infeasible for the given timeframe. Although Direct Connect is reliable, the data transfer will still take more than 2 weeks using only a 1Gb connection.

Overall Explanation:

Using a Snowball Edge Storage Optimized device combined with AWS DMS and AWS SCT provides a cost-effective solution for migrating a 20 TB database to AWS within 2 weeks with minimal downtime. AWS Snowmobile is overkill for the given data size, the Compute Optimized Snowball provides unnecessary compute capabilities, and AWS Direct Connect takes too long to set up. Using DMS with SCT ensures the database is migrated effectively while also allowing for replication of ongoing changes. This ensures the least amount of downtime.

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Question 5

Not Attempted

Databases



A company recently migrated its on-premises PostgreSQL database to an Amazon RDS for PostgreSQL DB instance. Following the successful launch of a new product, the database workload has significantly increased. The company aims to accommodate this larger workload without adding more infrastructure. Which solution will meet these requirements MOST cost-effectively?

- ☒ Purchase reserved DB instances sized for the total workload. Then, scale up the existing Amazon RDS for PostgreSQL DB instance to a larger instance type.

This option provides cost savings through reserved instances and scales vertically by increasing the instance size, avoiding the addition of new infrastructure.

- ☐ Convert the Amazon RDS for PostgreSQL DB instance to a Multi-AZ DB instance.

Multi-AZ primarily provides high availability and failover capabilities, not necessarily increased capacity for a larger workload. While it distributes load across multiple zones, it doesn't directly address the need for a larger instance size to handle the increased workload.

- ☐ Purchase reserved DB instances for the total workload. Add another Amazon RDS for PostgreSQL DB instance to distribute the workload.

This option involves adding another DB instance, which directly contradicts the requirement of "without adding infrastructure."

- ⊗ Convert the Amazon RDS for PostgreSQL DB instance to an on-demand DB instance.

While on-demand instances offer flexibility, they are generally more expensive than reserved instances for predictable workloads. Since the company knows the workload has increased, reserved instances would be more cost-effective.

Overall Explanation:

The most cost-effective solution that avoids adding infrastructure is to buy reserved DB instances to cover the workload and scale the existing RDS instance to a larger instance size. Multi-AZ is for High Availability, adding another instance adds infrastructure, and On-demand instances are more expensive for known workloads. Therefore, Option 1 is the optimal choice.

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Question 6

Not Attempted

Security



A company's ecommerce website runs on Amazon EC2 instances behind an Application Load Balancer (ALB) in an Auto Scaling group. The site's performance is suffering due to a high request rate from illegitimate external systems using rapidly changing IP addresses. The security team is concerned about potential DDoS attacks. The company needs to block these illegitimate requests while minimizing the impact on legitimate users. What should a solutions architect recommend?

- ⊗ Deploy Amazon Inspector and associate it with the ALB.

Amazon Inspector is a vulnerability management service that automates security assessments. It's not designed for real-time DDoS mitigation or IP-based request filtering.

- ✔ Deploy AWS WAF, associate it with the ALB, and configure a rate-limiting rule.

AWS WAF allows you to create rules to filter traffic based on various criteria, including IP addresses and request rates. Configuring a rate-limiting rule can effectively block excessive requests from specific IP addresses or ranges, mitigating the impact of a DDoS attack.

- ⊗ Deploy rules to the network ACLs associated with the ALB to block the incoming traffic.

Network ACLs operate at the subnet level (Layer 3 and 4) and are stateless, making them less suitable for complex filtering logic and rapidly changing IP addresses associated with a DDoS attack. Updating ACLs for rapidly changing IPs is not practical and can be disruptive.

- ⊗ Deploy Amazon GuardDuty and enable rate-limiting protection when configuring GuardDuty.

Amazon GuardDuty is a threat detection service that analyzes logs for malicious activity. It's not a tool for real-time traffic filtering or DDoS mitigation. GuardDuty doesn't have a feature to enable rate-limiting protection.

Overall Explanation:

AWS WAF, associated with the ALB, and configured with a rate-limiting rule, is the best solution for blocking illegitimate requests from changing IP addresses while minimizing the impact on legitimate users. WAF operates at the application layer and can effectively filter traffic based on various criteria, including IP addresses and request rates. While AWS Shield Advanced is the best protection, WAF is a viable alternative when Shield is not available. Network ACLs are not suitable for this scenario due to their limitations in handling rapidly changing IPs and complex filtering logic. Amazon Inspector and GuardDuty are not designed for real-time DDoS mitigation or request filtering.

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Question 7

Not Attempted

Security



A company needs to share accounting data with an external auditor. The data resides in an Amazon RDS DB instance within a private subnet. The auditor has their own AWS account and requires a separate copy of the database. What is the MOST secure method for the company to provide the database to the auditor?

- ⊗ Create a read replica of the database. Configure IAM standard database authentication to grant the auditor access.

Creating a read replica and granting access via IAM might expose the database to potential vulnerabilities. It requires more complex IAM configurations and opens the database directly to external access, which may not be the most secure approach.

- ⊗ Export the database contents to text files. Store the files in an Amazon S3 bucket. Create a new IAM user for the auditor. Grant the user access to the S3 bucket.

Exporting to text files exposes the data in an unencrypted format in S3, increasing the risk of unauthorized access or data breaches. Managing IAM users and bucket permissions adds complexity and potential security gaps.

- ⊗ Copy a snapshot of the database to an Amazon S3 bucket. Create an IAM user. Share the user's keys with the auditor to grant access to the object in the S3 bucket.

Copying a snapshot to S3, while potentially better than option B, still involves storing data outside of RDS without built-in encryption at rest. Sharing IAM user keys presents a security risk if compromised.

- ✔ Create an encrypted snapshot of the database. Share the snapshot with the auditor. Allow access to the AWS Key Management Service (AWS KMS) encryption key.

Creating an encrypted snapshot ensures data is protected at rest. Sharing the snapshot and KMS key allows the auditor to securely restore and access the data in their own account. This approach provides strong security and access control.

Overall Explanation:

The most secure method is to create an encrypted snapshot and share it with the auditor, along with access to the KMS key used for encryption. This ensures that the data is encrypted at rest and that only the authorized auditor can decrypt and access the data. This approach minimizes risks associated with data exposure during transfer and storage while allowing the auditor to create their own copy.

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Question 8

Not Attempted

Networking



A solutions architect configured a VPC with a limited range of IP addresses. The number of Amazon EC2 instances within the VPC is growing, leading to an insufficient number of IP addresses for future workloads. Which solution resolves this issue with the LEAST operational overhead?

- ✔ Add an additional IPv4 CIDR block to increase the number of IP addresses and create additional subnets in the VPC. Create new resources in the new subnets by using the new CIDR.

Adding a CIDR block is the simplest way to expand the IP range of a VPC without creating new infrastructure. It minimizes operational overhead by utilizing existing resources.

- ⊗ Create a second VPC with additional subnets. Use a peering connection to connect the second VPC with the first VPC. Update the routes and create new resources in the subnets of the second VPC.

VPC peering introduces operational overhead related to managing the peering connection, routing tables, and security groups across two VPCs.

- ⊗ Use AWS Transit Gateway to add a transit gateway and connect a second VPC with the first VPC. Update the routes of the transit gateway and VPCs. Create new resources in the subnets of the second VPC.

Transit Gateway provides more advanced routing capabilities for larger, more complex networks. Introducing Transit Gateway for this scenario would be overkill and increase operational complexity.

- ⊗ Create a second VPC. Create a Site-to-Site VPN connection between the first VPC and the second VPC by using a VPN-hosted solution on Amazon EC2 and a virtual private gateway. Update the route between VPCs to the traffic through the VPN. Create new resources in the subnets of the second VPC.

Creating a VPN connection adds significant overhead due to configuring and maintaining the VPN tunnel and the required networking components. It is far more complex than simply adding a CIDR.

Overall Explanation:

Adding an additional IPv4 CIDR block to the existing VPC and creating new subnets within it provides the necessary IP addresses with the least amount of operational overhead. All other options involve creating new VPCs and network infrastructure, increasing complexity and maintenance efforts. After the VPC has been created, you can associate additional IPv4 CIDR blocks with the VPC.

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Question 9

Not Attempted

Storage



A company stores critical data in an Amazon S3 bucket and needs to protect this data from accidental deletion. Which combination of actions should a solutions architect take to best meet these requirements? (Choose two.)

- ✔ Enable versioning on the S3 bucket.

Enabling versioning allows you to recover objects that were accidentally deleted or overwritten by keeping historical versions of the objects.

✓ Enable MFA Delete on the S3 bucket.

Enabling MFA Delete adds an extra layer of security by requiring multi-factor authentication for object deletion, preventing accidental or unauthorized deletions.

✗ Create a bucket policy on the S3 bucket.

Bucket policies primarily control access and permissions but do not directly prevent accidental deletion.

✗ Enable default encryption on the S3 bucket.

Default encryption protects data at rest but does not prevent accidental deletion.

✗ Create a lifecycle policy for the objects in the S3 bucket.

Lifecycle policies automate object management but do not directly protect against accidental deletion; they primarily focus on cost optimization and data retention.

Overall Explanation:

Enabling versioning allows for the recovery of accidentally deleted objects, while MFA Delete adds a layer of security requiring multi-factor authentication to confirm object deletion, thus preventing accidental or unauthorized deletions. Bucket policies, encryption, and lifecycle policies do not directly address accidental data deletion.

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Question 10

Not Attempted

Databases



A company performed application testing using an Amazon RDS for MySQL DB instance. Before terminating the DB instance, a solutions architect created two backups. The first backup was a database dump using the `mysqldump` utility. The second backup was created by enabling the final DB snapshot option during RDS termination. The company plans a new test cycle and wants to create a new DB instance using the most recent backup, hosted on a MySQL-compatible Amazon Aurora DB cluster. Which two solutions can create the new DB instance?

✓ Import the RDS snapshot directly into Aurora.

This is a valid approach. RDS snapshots can be directly imported into Aurora to create a new DB instance.

- ☐ Upload the RDS snapshot to Amazon S3. Then import the RDS snapshot into Aurora.
This is unnecessary. RDS snapshots are already stored within AWS infrastructure; directly importing them into Aurora is more efficient.
- ☒ Upload the database dump to Amazon S3. Then import the database dump into Aurora.
This is a valid approach. The database dump can be uploaded to S3 and then imported into Aurora to create a new DB instance.
- ☐ Use AWS Database Migration Service (AWS DMS) to import the RDS snapshot into Aurora.
AWS DMS is not required to migrate RDS snapshots to Aurora. Direct import is possible.
- ☐ Upload the database dump to Amazon S3. Then use AWS Database Migration Service (AWS DMS) to import the database dump into Aurora.
While DMS can be used, it is not required for migrating a MySQL dump to Aurora. Importing the dump directly is a simpler approach.

Overall Explanation:

The two correct solutions are importing the RDS snapshot directly into Aurora and uploading the database dump to Amazon S3, then importing the database dump into Aurora. RDS snapshots can be directly imported to create a new Aurora instance. Database dumps created with `mysqldump` can be uploaded to S3 and then imported into Aurora. Using DMS is an option but not necessary, and uploading the snapshot to S3 is redundant as the snapshots are already present in AWS.

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Question 11

Not Attempted

[Compute](#)

A company hosts a multi-tier web application on Amazon Linux Amazon EC2 instances behind an Application Load Balancer. The instances run in an Auto Scaling group across multiple Availability Zones. The company observes that the Auto Scaling group launches more On-Demand Instances when the application's end users access high volumes of static web content. The company wants to optimize cost. What should a solutions architect do to redesign the application MOST cost-effectively?

- ⊗ Update the Auto Scaling group to use Reserved Instances instead of On-Demand Instances.

Using Reserved Instances might reduce costs for the instances that are always running but doesn't address the issue of scaling due to static content requests.

- ⊗ Update the Auto Scaling group to scale by launching Spot Instances instead of On-Demand Instances.

Using Spot Instances can reduce costs but introduces the risk of instances being terminated, which might impact application availability, especially when handling high traffic.

- ✓ Create an Amazon CloudFront distribution to host the static web contents from an Amazon S3 bucket.

Offloading static content to Amazon CloudFront and S3 reduces the load on the EC2 instances, which will reduce the need to scale the Auto Scaling group for static content requests, therefore it is the most cost-effective option.

- ⊗ Create an AWS Lambda function behind an Amazon API Gateway API to host the static website contents.

Using Lambda and API Gateway to serve static content is generally more expensive and complex than using CloudFront and S3.

Overall Explanation:

Using Amazon CloudFront with an Amazon S3 bucket is the most cost-effective solution for serving static content. This approach reduces the load on the EC2 instances by caching the content at edge locations, which reduces the number of On-Demand Instances required. Reserved or Spot Instances do not directly address the problem of serving static content, and Lambda/API Gateway is less cost-effective than CloudFront/S3 for this purpose.

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Question 12

Not Attempted

Security



A company stores several petabytes of data across multiple AWS accounts and uses AWS Lake Formation to manage its data lake. The company's data science team needs to securely share selective data from its accounts with the company's engineering team for analytical purposes, minimizing operational overhead.

- ☐ Copy the required data to a common account. Create an IAM access role in that account. Grant access by specifying a permission policy that includes users from the engineering team accounts as trusted entities.
Copying data involves significant overhead and complexity in managing data duplication and synchronization.
- ☐ Use the Lake Formation permissions Grant command in each account where the data is stored to allow the required engineering team users to access the data.
While granular, configuring permissions in each account increases operational overhead.
- ☐ Use AWS Data Exchange to privately publish the required data to the required engineering team accounts.
Data Exchange is more suitable for external data sharing and introduces unnecessary overhead for internal teams.
- ☒ Use Lake Formation tag-based access control to authorize and grant cross-account permissions for the required data to the engineering team accounts.
Tag-based access control allows granular, centralized control with minimal operational overhead. It facilitates cross-account permissions based on tags.

Overall Explanation:

Lake Formation's tag-based access control provides a centralized and scalable solution for securely sharing data across accounts. It allows you to define tags and tag-based policies to grant selective access to the required data for the engineering team accounts without copying the data or managing permissions individually in each account. This approach minimizes operational overhead compared to other options.

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Question 13

Not Attempted

[Storage](#)

A company is developing a globally accessible web application on AWS. Users from various geographic locations will upload and download unique data files, some reaching gigabytes in size. The development team requires a cost-effective solution that minimizes upload/download latency and maximizes performance for these large file transfers. Which approach should a solutions architect recommend?

✔ Use Amazon S3 with Transfer Acceleration to host the application.

S3 Transfer Acceleration is designed to optimize the transfer of large files over long distances by utilizing CloudFront's globally distributed edge locations. It's cost-effective for accelerating uploads and downloads of unique data.

✘ Use Amazon S3 with CacheControl headers to host the application.

CacheControl headers are primarily used for managing browser caching and are not effective for optimizing large file uploads or downloads. They are not designed for unique data.

✘ Use Amazon EC2 with Auto Scaling and Amazon CloudFront to host the application.

While EC2 with Auto Scaling and CloudFront can handle web application traffic and caching of static content, it doesn't directly address the optimization of large, unique file uploads. Where will the files be stored?

✘ Use Amazon EC2 with Auto Scaling and Amazon ElastiCache to host the application.

ElastiCache is an in-memory data store, suitable for caching frequently accessed data, not for handling large file uploads and downloads. Auto Scaling with EC2 is irrelevant to the problem.

Overall Explanation:

Amazon S3 Transfer Acceleration is the most appropriate solution for minimizing upload and download latency of large, unique files across geographic regions. It leverages CloudFront's edge locations to accelerate transfers. The problem focuses on upload and download, not hosting. The question specifically refers to unique data up to gigabytes, this invalidates caching solutions like CloudFront because caching is effective for repeated requests, not unique data.

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Question 14

Not Attempted

Databases



A company has hired a solutions architect to design a reliable architecture for its application. The application consists of one Amazon RDS DB instance and two manually provisioned Amazon EC2 instances that run web servers. The EC2 instances are located in a single Availability Zone. An employee recently deleted the DB instance, and the application was unavailable for 24 hours as a result. The company is concerned with the overall reliability of its

environment. What should the solutions architect do to maximize reliability of the application's infrastructure?

- ⊗ Delete one EC2 instance and enable termination protection on the other EC2 instance. Update the DB instance to be Multi-AZ, and enable deletion protection.

Deleting one EC2 instance reduces capacity and doesn't address the lack of redundancy across Availability Zones. While enabling termination protection on the remaining instance prevents accidental deletion, it doesn't solve the problem of a single point of failure. Although it takes care of reliability of DB instance, it does not address the reliability of the EC2 instances.

- ✓ Update the DB instance to be Multi-AZ, and enable deletion protection. Place the EC2 instances behind an Application Load Balancer, and run them in an EC2 Auto Scaling group across multiple Availability Zones.

This option provides the best solution for high availability and fault tolerance. Multi-AZ RDS protects against DB instance failure and data loss. The Auto Scaling group and Application Load Balancer distribute traffic across multiple Availability Zones, ensuring that the application remains available even if one or more instances fail. The deletion protection prevents the DB from accidental deletion. This option addresses both EC2 and DB reliability.

- ⊗ Create an additional DB instance along with an Amazon API Gateway and an AWS Lambda function. Configure the application to invoke the Lambda function through API Gateway. Have the Lambda function write the data to the two DB instances.

This option introduces unnecessary complexity with API Gateway and Lambda. While it provides redundancy at the data layer, it doesn't address the immediate issue of RDS deletion protection and isn't as efficient as Multi-AZ RDS. The DB instance's reliability is not taken care of using Multi-AZ.

- ⊗ Place the EC2 instances in an EC2 Auto Scaling group that has multiple subnets located in multiple Availability Zones. Use Spot Instances instead of On-Demand Instances. Set up Amazon CloudWatch alarms to monitor the health of the instances Update the DB instance to be Multi-AZ, and enable deletion protection.

While using an Auto Scaling group across multiple AZs improves EC2 instance reliability, using Spot Instances can introduce instability due to potential interruptions. While it appears to address EC2 and DB reliability, it introduces potential cost savings over reliability (Spot Instances).

Overall Explanation:

Option B is the most suitable solution. Updating the DB instance to Multi-AZ with deletion protection prevents data loss and service interruption due to accidental deletion or hardware failure. Placing the EC2 instances behind an Application Load Balancer and running them in an Auto Scaling group across multiple Availability Zones ensures high availability and fault tolerance for the web server tier. Option A doesn't provide high

availability for EC2 instances. Option C introduces unnecessary complexity. Option D uses Spot Instances, which could lead to interruptions, prioritizing cost savings over consistent reliability. Therefore, option B addresses both the database and web server reliability concerns most effectively.

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Question 15

Not Attempted

Storage



A company is storing 700 terabytes of data on a large network-attached storage (NAS) system in its corporate data center. The company has a hybrid environment with a 10 Gbps AWS Direct Connect connection. After an audit from a regulator, the company has 90 days to move the data to the cloud. The company needs to move the data efficiently and without disruption. The company still needs to be able to access and update the data during the transfer window. Which solution will meet these requirements?

- ☒ Create an AWS DataSync agent in the corporate data center. Create a data transfer task. Start the transfer to an Amazon S3 bucket.

AWS DataSync efficiently transfers large datasets from on-premises NAS to Amazon S3 over Direct Connect while allowing continued access and updates to the data during the transfer process.

- ☐ Back up the data to AWS Snowball Edge Storage Optimized devices. Ship the devices to an AWS data center. Mount a target Amazon S3 bucket on the on-premises file system. Snowball Edge Storage Optimized devices have a limited capacity and may require multiple devices for 700 TB of data. More importantly, this solution does not provide continuous data access and updates during the transfer window.

- ☐ Use rsync to copy the data directly from local storage to a designated Amazon S3 bucket over the Direct Connect connection. While rsync can be used for data transfer, it may not be as efficient or robust as AWS DataSync for large datasets and doesn't inherently offer the same level of management and monitoring capabilities.

- ☐ Back up the data on tapes. Ship the tapes to an AWS data center. Mount a target Amazon S3 bucket on the on-premises file system. Using tapes for backup and transfer is slow and inefficient and doesn't allow accessing/updating data during the transfer. Also, mounting an S3 bucket on-premises using tapes is not a standard practice.

Overall Explanation:

AWS DataSync is the most suitable solution because it allows efficient and secure transfer of 700 TB of data to Amazon S3 over Direct Connect within the given timeframe, without disruption and while maintaining data accessibility and updatability during the transfer window. Snowball Edge devices might be limited by their capacity, and rsync might not be as efficient. Tape backups are slow and not ideal for this scenario.

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Question 16

Not Attempted

Storage



A company stores data in PDF format in an Amazon S3 bucket. The company must adhere to a legal requirement to retain all new and existing data in Amazon S3 for 7 years. Which solution will meet these requirements with the LEAST operational overhead?

- ☐ Turn on the S3 Versioning feature for the S3 bucket. Configure S3 Lifecycle to delete the data after 7 years. Configure multi-factor authentication (MFA) delete for all S3 objects.
S3 Versioning and Lifecycle policies don't enforce immutability for a specific retention period. MFA delete adds a layer of security, but doesn't guarantee retention.
- ☐ Turn on S3 Object Lock with governance retention mode for the S3 bucket. Set the retention period to expire after 7 years. Recopy all existing objects to bring the existing data into compliance.
Governance retention mode allows users with specific IAM permissions to alter or delete objects. Recopying adds significant operational overhead.
- ☐ Turn on S3 Object Lock with compliance retention mode for the S3 bucket. Set the retention period to expire after 7 years. Recopy all existing objects to bring the existing data into compliance.
Compliance retention mode enforces immutability, but manually recopying all existing objects introduces substantial operational overhead.
- ☒ Turn on S3 Object Lock with compliance retention mode for the S3 bucket. Set the retention period to expire after 7 years. Use S3 Batch Operations to bring the existing data into compliance.

Compliance retention mode enforces immutability. S3 Batch Operations automates the application of Object Lock to existing objects, minimizing operational overhead.

Overall Explanation:

S3 Object Lock in compliance mode ensures data retention for the specified period. Using S3 Batch Operations is the most efficient method to apply this setting to existing objects, thus meeting the requirements with the least operational burden. Versioning is not relevant here. Governance mode does not enforce Object Lock, and manual recopying of existing objects is operationally expensive. S3 Batch Operations reduces manual overhead and risk.

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Question 17

Not Attempted

Networking



A company has a stateless web application that runs on AWS Lambda functions invoked by Amazon API Gateway. The company aims to deploy the application across multiple AWS Regions to provide Regional failover capabilities. Which action should a solutions architect take to route traffic to multiple Regions?

- ☒ Create Amazon Route 53 health checks for each Region. Use an active-active failover configuration.

This is the correct answer. Route 53 with health checks and an active-active failover configuration allows for monitoring endpoint health in each region and automatically routing traffic to healthy endpoints, ensuring failover capabilities.

- ☐ Create an Amazon CloudFront distribution with an origin for each Region. Use CloudFront health checks to route traffic.

CloudFront is primarily a CDN and while it can distribute traffic to multiple origins, it is not designed for automatic failover between regions based on health checks like Route 53. It's more focused on caching and reducing latency.

- ☐ Create a transit gateway. Attach the transit gateway to the API Gateway endpoint in each Region. Configure the transit gateway to route requests.

Transit Gateway is designed for VPC connectivity and not for routing external requests to API Gateway endpoints. It's not the appropriate service for regional failover of a web application.

- ⊗ Create an Application Load Balancer in the primary Region. Set the target group to point to the API Gateway endpoint hostnames in each Region.

This option creates a dependency on a primary region. If the primary region hosting the ALB fails, the failover mechanism will not work. Additionally, an ALB typically resides within a single region.

Overall Explanation:

Amazon Route 53 is the most suitable solution for routing traffic to multiple AWS Regions and providing regional failover for a stateless web application. By utilizing health checks and an active-active failover configuration, Route 53 can monitor the health of endpoints in each Region and automatically redirect traffic to healthy endpoints in other Regions in case of a failure. This ensures high availability and resilience. CloudFront is primarily for caching, Transit Gateway for VPC connectivity, and an ALB in a single region introduces a single point of failure.

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Question 18

Not Attempted

Networking



A company has two VPCs named Management and Production. The Management VPC uses VPNs through a customer gateway to connect to a single device in the data center. The Production VPC uses a virtual private gateway with two attached AWS Direct Connect connections. The Management and Production VPCs both use a single VPC peering connection to allow communication between the applications. What should a solutions architect do to mitigate any single point of failure in this architecture?

- ⊗ Add a set of VPNs between the Management and Production VPCs.

This option does not address the single point of failure in the connection between the Management VPC and the data center.

- ⊗ Add a second virtual private gateway and attach it to the Management VPC.

A virtual private gateway is used for Direct Connect and VPN connections, but in this case, the customer gateway device in the data center is the single point of failure.

- ✓ Add a second set of VPNs to the Management VPC from a second customer gateway device.

This is the correct answer. Adding a second VPN connection through a second customer gateway device provides redundancy and eliminates the single point of failure in the connection between the Management VPC and the data center.

- ⊗ Add a second VPC peering connection between the Management VPC and the Production VPC.

Adding a second VPC peering connection addresses the VPC peering connection between Management and Production VPCs, which isn't the single point of failure focused on in the question.

Overall Explanation:

The Management VPC currently uses VPNs through a single customer gateway to connect to the data center. To mitigate this single point of failure, adding a second set of VPNs to the Management VPC from a second customer gateway device is the best solution. This provides redundancy, ensuring that connectivity is maintained even if one customer gateway fails.

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Question 19

Not Attempted

Databases



A company currently operates its application on an Oracle database. Due to limited resources for database management, backup administration, and data center upkeep, the company wants to migrate to AWS quickly. The application utilizes third-party database features that necessitate privileged access. Which solution offers the MOST cost-effective migration path to AWS?

- ⊗ Migrate the database to Amazon RDS for Oracle. Replace third-party features with cloud services.

This option may not be feasible if replacing the third-party features involves significant development effort or if equivalent cloud services are unavailable. It doesn't address the need for privileged access directly.

- ✔ Migrate the database to Amazon RDS Custom for Oracle. Customize the database settings to support third-party features.

Amazon RDS Custom allows for customization of the underlying database and operating system, enabling support for third-party features requiring privileged access. It also retains the benefits of RDS automation.

- ⊗ Migrate the database to an Amazon EC2 Amazon Machine Image (AMI) for Oracle.

Customize the database settings to support third-party features.

While EC2 provides full control, it shifts the responsibility for database administration, backup, and maintenance to the company, negating the initial goal of reducing resource burden and potentially increasing costs.

- ⊗ Migrate the database to Amazon RDS for PostgreSQL by rewriting the application code to remove dependency on Oracle APEX.

Rewriting the application code is a time-consuming and expensive endeavor, making it the least cost-effective option. It also doesn't align with the requirement for a quick migration.

Overall Explanation:

Amazon RDS Custom for Oracle is the most suitable solution because it provides a managed database service while allowing the necessary customization to support the third-party features requiring privileged access. This balances cost-effectiveness with the need to maintain application functionality without extensive code changes or increased administrative overhead. RDS Custom provides the automation benefits of RDS while offering the necessary access for legacy and custom applications. Migrating to EC2 would require more management overhead and rewriting the application for PostgreSQL would be extremely costly.

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Question 20

Not Attempted

Serverless



A company has a data ingestion workflow that consists of the following:

- An Amazon Simple Notification Service (Amazon SNS) topic for notifications about new data deliveries
- An AWS Lambda function to process the data and record metadata

The company observes that the ingestion workflow fails occasionally because of network connectivity issues. When such a failure occurs, the Lambda function does not ingest the corresponding data unless the company manually reruns the job. Which combination of actions should a solutions architect take to ensure that the Lambda function ingests all data in the future? (Choose two.)

- ⊗ Deploy the Lambda function in multiple Availability Zones.

Deploying the Lambda function in multiple Availability Zones improves availability and fault tolerance but does not guarantee ingestion of all data if the network connectivity issue prevents the

function from being triggered in the first place.

- ✔ Create an Amazon Simple Queue Service (Amazon SQS) queue, and subscribe it to the SNS topic.

Creating an SQS queue and subscribing it to the SNS topic ensures that messages are reliably stored, even during network connectivity issues. This decouples the notification and processing layers.

- ✘ Increase the CPU and memory that are allocated to the Lambda function.

Increasing CPU and memory allocation may improve Lambda function performance but does not address the root cause of network connectivity failures leading to message loss.

- ✘ Increase provisioned throughput for the Lambda function.

Lambda functions are automatically scaled by AWS, and provisioned throughput is not configurable. This option is therefore invalid.

- ✔ Modify the Lambda function to read from an Amazon Simple Queue Service (Amazon SQS) queue.

Modifying the Lambda function to read from the SQS queue ensures that messages are processed asynchronously and are not lost even if the Lambda function fails temporarily due to network connectivity or other issues. The Lambda function can retry processing of the messages from the queue.

Overall Explanation:

To ensure that the Lambda function ingests all data, even during network connectivity issues, it's important to decouple the notification and processing layers using SQS. An SQS queue subscribed to the SNS topic will reliably store messages during outages (Option B). Modifying the Lambda function to read from the SQS queue ensures that messages are eventually processed (Option E), even if the initial attempt fails. Options A, C, and D do not directly address the issue of messages being lost due to network connectivity problems.

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Question 21

Not Attempted

Compute



A company hosts a three-tier web application on a single server. The company plans to migrate the application to the AWS Cloud, aligning with the AWS Well-Architected Framework and

adhering to AWS recommended best practices for security, scalability, and resiliency. Which combination of solutions will meet these requirements?

- ☐ Create a VPC across two Availability Zones with the application's existing architecture. Host the application with the existing architecture on an Amazon EC2 instance in a private subnet in each Availability Zone with EC2 Auto Scaling groups. Secure the EC2 instance with security groups and network access control lists (network ACLs).
This option is not ideal because it suggests using the existing architecture, which might not be optimized for the cloud. While it utilizes Auto Scaling groups and security measures, it does not fully address the scalability and reliability of each tier.
- ☐ Set up security groups and network access control lists (network ACLs) to control access to the database layer. Set up a single Amazon RDS database in a private subnet.
Using a single RDS instance does not align with high availability best practices, a key component of the Well-Architected Framework.
- ☒ Create a VPC across two Availability Zones. Refactor the application to host the web tier, application tier, and database tier. Host each tier on its own private subnet with Auto Scaling groups for the web tier and application tier.
This option is a good starting point for refactoring the application and provides a more scalable and resilient design. Separating tiers into different subnets enhances security and allows independent scaling.
- ☐ Use a single Amazon RDS database. Allow database access only from the application tier security group.
This option is not suitable because relying on a single RDS instance does not ensure high availability.
- ☒ Use Elastic Load Balancers in front of the web tier. Control access by using security groups containing references to each layer's security groups.
Using Elastic Load Balancers (ELBs) allows distribution of traffic across multiple web tier instances, improving scalability and availability. Security groups provide granular control over network traffic.
- ☒ Use an Amazon RDS database Multi-AZ cluster deployment in private subnets. Allow database access only from application tier security groups.
Deploying an Amazon RDS database in a Multi-AZ configuration offers high availability and automatic failover. Placing the database in private subnets improves security, and restricting access to the application tier follows the principle of least privilege.

Overall Explanation:

The best combination of solutions to meet the company's requirements for migration to the AWS Cloud, adhering to the AWS Well-Architected Framework, and ensuring security, scalability, and resiliency, are options C, E, and F. Option C supports scalability and resilience by separating tiers. Option E enhances availability through Elastic Load Balancers, and option F guarantees database high availability and secures access to the database.

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Question 22

Not Attempted

[Compute](#)

A company is migrating its applications and databases to the AWS Cloud. The company will use Amazon Elastic Container Service (Amazon ECS), AWS Direct Connect, and Amazon RDS. Which activities will be managed by the company's operational team? (Choose three.)

- ☐ Management of the Amazon RDS infrastructure layer, operating system, and platforms
AWS manages the underlying infrastructure of RDS.
- ☒ Creation of an Amazon RDS DB instance and configuring the scheduled maintenance window
The company is responsible for creating and managing RDS instances, including maintenance windows.
- ☒ Configuration of additional software components on Amazon ECS for monitoring, patch management, log management, and host intrusion detection
The company is responsible for configuring monitoring, logging, and security on their ECS containers.
- ☐ Installation of patches for all minor and major database versions for Amazon RDS
AWS handles patching of the RDS infrastructure.
- ☐ Ensure the physical security of the Amazon RDS infrastructure in the data center
AWS handles the physical security of the data center where RDS resides.
- ☒ Encryption of the data that moves in transit through Direct Connect
The company is responsible for encrypting data in transit across Direct Connect.

Overall Explanation:

The company is responsible for creating RDS instances (B), configuring monitoring and security for ECS containers (C), and encrypting data traversing Direct Connect (F). AWS manages the underlying infrastructure and physical security of RDS, and handles patching. Direct Connect doesn't automatically encrypt the connection, the company is responsible.

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Question 23

Not Attempted

Serverless



A company runs a Java-based job on an Amazon EC2 instance. The job runs every hour and takes 10 seconds to run. The job runs on a scheduled interval and consumes 1 GB of memory. The CPU utilization of the instance is low except for short surges during which the job uses the maximum CPU available. The company wants to optimize the costs to run the job. Which solution will meet these requirements?

- ☐ Use AWS App2Container (A2C) to containerize the job. Run the job as an Amazon Elastic Container Service (Amazon ECS) task on AWS Fargate with 0.5 virtual CPU (vCPU) and 1 GB of memory.

While Fargate is serverless, it is not as cost-effective for a short-running, infrequent job as Lambda. You are still paying for the resource even when it is not running.

- ☒ Copy the code into an AWS Lambda function that has 1 GB of memory. Create an Amazon EventBridge scheduled rule to run the code each hour.

This is the most cost-effective solution. Lambda allows you to run code without provisioning or managing servers. You only pay for the compute time you consume. EventBridge can schedule the execution.

- ☐ Use AWS App2Container (A2C) to containerize the job. Install the container in the existing Amazon Machine Image (AMI). Ensure that the schedule stops the container when the task finishes.

This solution still requires the EC2 instance to be running, even if stopped and started, which incurs costs. Lambda provides a more cost-effective approach.

- ☐ Configure the existing schedule to stop the EC2 instance at the completion of the job and restart the EC2 instance when the next job starts.

Stopping and starting an EC2 instance can take time, potentially exceeding the 10-second job runtime. It also requires infrastructure management, which Lambda avoids.

Overall Explanation:

Lambda functions combined with EventBridge scheduled rules provide a cost-effective, easily managed, and scalable solution for running short, scheduled jobs. Lambda minimizes the need to manage infrastructure because you only pay for the compute time used during job execution. Lambda allows you to allocate memory for your functions in increments of 1 MB, ranging from a minimum of 128 MB to a maximum of 10,240 MB (10 GB). Option B correctly leverages Lambda and EventBridge to optimize costs.

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Question 24

Not Attempted

Storage



A company needs to establish a backup strategy for Amazon EC2 instances and multiple Amazon S3 buckets. Due to strict regulatory compliance, the company is required to retain all backup files for a predefined period. Critically, the backed-up files must remain unaltered throughout this retention period. Which solution effectively meets these requirements and ensures immutability during the retention period?

- ☐ Use AWS Backup to create a backup vault that has a vault lock in governance mode. Create the required backup plan.
Governance mode in AWS Backup Vault Lock allows certain privileged users to alter or delete backups, which violates the requirement that files must not be altered for the duration of the retention period.
- ☐ Use Amazon Data Lifecycle Manager to create the required automated snapshot policy. Amazon Data Lifecycle Manager is primarily used for managing the lifecycle of EBS snapshots and does not inherently provide immutability or a vault lock feature for regulatory compliance regarding backup alteration.
- ☐ Use Amazon S3 File Gateway to create the backup. Configure the appropriate S3 Lifecycle management. Amazon S3 File Gateway primarily focuses on providing on-premises applications access to S3 storage. While S3 Lifecycle management can help manage storage costs, it doesn't offer a native mechanism to ensure immutability with a strict compliance lock.
- ☒ Use AWS Backup to create a backup vault that has a vault lock in compliance mode. Create the required backup plan.

Compliance mode in AWS Backup Vault Lock enforces immutability, preventing any alterations or deletions of the backups during the specified retention period. This aligns directly with the regulatory requirement to prevent any changes to the backup files.

Overall Explanation:

The correct answer is ****D****. AWS Backup with Vault Lock in compliance mode ensures that once backups are created, they cannot be altered or deleted for the duration of the retention period. This satisfies the requirement for data immutability dictated by the regulatory requirements. Governance mode does not provide the immutability needed. Data Lifecycle Manager manages snapshots and S3 File Gateway with lifecycle policies do not directly address the compliance requirements of locking backup files.

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Question 25

Not Attempted

[Automation](#)

A company's resources are spread across multiple AWS Regions and accounts. A newly hired solutions architect discovers that the previous employee did not document the resource inventory. The solutions architect needs to build and map the relationship details of the various workloads across all accounts in the MOST operationally efficient way. Which solution meets these requirements?

- ☐ Use AWS Systems Manager Inventory to generate a map view from the detailed view report.

AWS Systems Manager Inventory primarily collects metadata about instances. It is not designed for generating comprehensive architecture diagrams of workloads across multiple accounts and Regions.

- ☐ Use AWS Step Functions to collect workload details. Build architecture diagrams of the workloads manually.

While Step Functions can orchestrate collecting workload details, manually building architecture diagrams is time-consuming and not operationally efficient.

- ☒ Use Workload Discovery on AWS to generate architecture diagrams of the workloads.

Workload Discovery on AWS is specifically designed for automatically discovering and mapping relationships between AWS resources across multiple accounts and Regions, providing visual architecture diagrams. This makes it the MOST operationally efficient option.

- ☐ Use AWS X-Ray to view the workload details. Build architecture diagrams with relationships.

AWS X-Ray is primarily used for application debugging and tracing requests. It's not designed for mapping out the entire architecture and relationships between resources across multiple accounts.

Overall Explanation:

Workload Discovery on AWS is a purpose-built tool that automatically generates visual mappings of architectures across AWS accounts and Regions. It discovers resources and their relationships, providing an accurate and up-to-date representation of the infrastructure. This automation makes it the MOST operationally efficient way to fulfill the requirement of building and mapping workload details.

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Question 26

Not Attempted

Monitoring



A company uses AWS Organizations. The company wants to operate some of its AWS accounts with different budgets. The company wants to receive alerts and automatically prevent provisioning of additional resources on AWS accounts when the allocated budget threshold is met during a specific period. Which combination of solutions will meet these requirements? (Choose three.)

- ☐ Use AWS Budgets to create a budget. Set the budget amount under the Cost and Usage Reports section of the required AWS accounts.

Setting budget amount under Cost and Usage Reports is not the correct approach. AWS Budgets are configured directly via the Billing and Cost Management console.

- ☒ Use AWS Budgets to create a budget. Set the budget amount under the Billing dashboards of the required AWS accounts.

AWS Budgets are set up within the AWS Billing and Cost Management console, specifically within the Budgets section.

- ☐ Create an IAM user for AWS Budgets to run budget actions with the required permissions.

Creating an IAM user is not recommended for AWS Budgets. It's better to create an IAM role.

- ☒ Create an IAM role for AWS Budgets to run budget actions with the required permissions.
An IAM role is required to grant AWS Budgets the necessary permissions to perform automated actions, like preventing resource provisioning.
- ☐ Add an alert to notify the company when each account meets its budget threshold. Add a budget action that selects the IAM identity created with the appropriate config rule to prevent provisioning of additional resources.
Budget actions should select an IAM identity with an appropriate service control policy (SCP). Config Rules can remediate resources but not stop the provisioning. SCPs are used to centrally control the AWS services that your organization can access.
- ☒ Add an alert to notify the company when each account meets its budget threshold. Add a budget action that selects the IAM identity created with the appropriate service control policy (SCP) to prevent provisioning of additional resources.
Adding alerts for budget thresholds and using a budget action that leverages an IAM identity with an appropriate SCP is the correct approach to prevent resource provisioning.

Overall Explanation:

To meet the requirements, you need to utilize AWS Budgets, IAM roles, and Service Control Policies (SCPs). Option B correctly sets up AWS Budgets via the Billing and Cost Management dashboard. Option D specifies the use of an IAM role, which provides the necessary permissions for AWS Budgets to perform actions. Option F combines alerts for budget thresholds with budget actions that utilize an IAM identity linked to an SCP, allowing for the prevention of resource provisioning when the budget is exceeded. Thus, BDF is the correct answer.

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Question 27

Not Attempted

[Compute](#)

A company runs applications on Amazon EC2 instances in one AWS Region. The company wants to back up the EC2 instances to a second Region. The company also wants to provision EC2 resources in the second Region and manage the EC2 instances centrally from one AWS account. Which solution will meet these requirements MOST cost-effectively?

- ⊗ Create a disaster recovery (DR) plan that has a similar number of EC2 instances in the second Region. Configure data replication.

This option involves maintaining a similar number of EC2 instances in the second region and configuring data replication. This will be costlier compared to using AWS Backup.

- ⊗ Create point-in-time Amazon Elastic Block Store (Amazon EBS) snapshots of the EC2 instances. Copy the snapshots to the second Region periodically.

While EBS snapshots can be used for backups, manually managing and copying them across regions adds complexity and can be more expensive than AWS Backup, especially when considering automation and central management.

- ✓ Create a backup plan by using AWS Backup. Configure cross-Region backup to the second Region for the EC2 instances.

AWS Backup allows for centralized management and automated cross-Region backups of EC2 instances. This is the most cost-effective approach for backing up and managing EC2 instances in another region.

- ⊗ Deploy a similar number of EC2 instances in the second Region. Use AWS DataSync to transfer the data from the source Region to the second Region.

Deploying a similar number of EC2 instances and using AWS DataSync to transfer data will be more expensive than using AWS Backup. This solution is not cost-effective.

Overall Explanation:

AWS Backup provides a cost-effective solution for backing up EC2 instances to a second Region, enabling central management from one AWS account. It also allows for cross-region backups. Therefore, using AWS Backup to create a backup plan and configuring cross-Region backup for the EC2 instances is the most cost-effective solution.

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Question 28

Not Attempted

Security



A company that uses AWS is building an application to transfer data to a product manufacturer. The company has its own identity provider (IdP). The company wants the IdP to authenticate application users while the users use the application to transfer data. The company must use Applicability Statement 2 (AS2) protocol. Which solution will meet these requirements?

- ⊗ Use AWS DataSync to transfer the data. Create an AWS Lambda function for IdP authentication.

AWS DataSync does not natively support the AS2 protocol.

- ⊗ Use Amazon AppFlow flows to transfer the data. Create an Amazon Elastic Container Service (Amazon ECS) task for IdP authentication.

Amazon AppFlow does not natively support the AS2 protocol.

- ✔ Use AWS Transfer Family to transfer the data. Create an AWS Lambda function for IdP authentication.

AWS Transfer Family supports the AS2 protocol for secure file transfers. AWS Lambda can be used to integrate with the company's IdP for authentication.

- ⊗ Use AWS Storage Gateway to transfer the data. Create an Amazon Cognito identity pool for IdP authentication.

While AWS Storage Gateway facilitates data transfer, it does not directly support the AS2 protocol. Amazon Cognito can handle authentication, but the data transfer requirement isn't met.

Overall Explanation:

AWS Transfer Family provides managed, secure file transfers using the AS2 protocol. AWS Lambda can be used to authenticate users with the company's existing IdP, providing a solution that satisfies both the data transfer and authentication requirements. Therefore, option C is the correct answer.

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Question 29

Not Attempted

Serverless



A solutions architect is designing a REST API in Amazon API Gateway for a cash payback service. The application requires 1 GB of memory and 2 GB of storage for its computation resources and that the data is in a relational format. Which additional combination of AWS services will meet these requirements with the LEAST administrative effort? (Choose two.)

⊗ Amazon EC2

Amazon EC2 provides compute capacity in the cloud, but requires more administrative overhead compared to serverless options. While it can fulfill the requirements, it's not the option with the least administrative effort.

✓ AWS Lambda

AWS Lambda provides a serverless compute service, which minimizes administrative overhead. It can be configured with 1 GB of memory, meeting the application requirements.

✓ Amazon RDS

Amazon RDS (Relational Database Service) provides managed relational databases, meeting the requirement for relational data storage.

⊗ Amazon DynamoDB

Amazon DynamoDB is a NoSQL database service and does not meet the relational data storage requirement.

⊗ Amazon Elastic Kubernetes Services (Amazon EKS)

Amazon EKS is a managed Kubernetes service. While it can be used, it involves significantly more administrative overhead compared to Lambda for this specific use case.

Overall Explanation:

AWS Lambda provides a serverless compute environment that is easily integrated with API Gateway and requires minimal administration. Amazon RDS offers managed relational database services and fulfills the need for relational data storage and the 2 GB storage requirement. Therefore, using Lambda for the compute and RDS for the relational data storage provides the solution with the least administrative effort.

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Question 30

Not Attempted

Monitoring



A company utilizes AWS Organizations to manage workloads across multiple AWS accounts. They have implemented a tagging policy that automatically applies department tags to AWS resources during tag creation. The accounting team needs to analyze Amazon EC2 spending, attributing costs to specific departments regardless of the AWS account where the resources reside. They have access to AWS Cost Explorer for all accounts within the organization and need

a streamlined approach to access relevant reports. Which solution offers the MOST operationally efficient way to meet these requirements?

- ☒ From the Organizations management account billing console, activate a user-defined cost allocation tag named department. Create one cost report in Cost Explorer grouping by tag name, and filter by EC2.

This is the correct answer. Activating the user-defined cost allocation tag from the management account allows for cost tracking across all accounts. Cost Explorer can then generate a single report, grouped by the 'department' tag and filtered for EC2 usage, providing the required departmental spending breakdown.

- ☐ From the Organizations management account billing console, activate an AWS-defined cost allocation tag named department. Create one cost report in Cost Explorer grouping by tag name, and filter by EC2.

AWS-defined tags are managed and defined by AWS, not the user. Since the company is using their own 'department' tags, this is not the right approach.

- ☐ From the Organizations member account billing console, activate a user-defined cost allocation tag named department. Create one cost report in Cost Explorer grouping by the tag name, and filter by EC2.

Activating the tag from a member account will not enable organization-wide cost allocation tracking. Cost allocation tags must be activated in the management account.

- ☐ From the Organizations member account billing console, activate an AWS-defined cost allocation tag named department. Create one cost report in Cost Explorer grouping by tag name, and filter by EC2.

This option combines the disadvantages of options B and C: it attempts to use an AWS-defined tag (which is not what the company uses) and activates it from a member account (which doesn't provide organization-wide visibility).

Overall Explanation:

The most operationally efficient solution is to activate the user-defined cost allocation tag ('department') from the Organizations management account billing console. This allows Cost Explorer to generate a consolidated report, grouped by the department tag and filtered by EC2 usage, providing the accounting team with the necessary information across all AWS accounts. Activating in the Management account ensures that the cost allocation applies across the whole organization.

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Question 31

Not Attempted

Security



A company provides marketing services to stores based on customer transaction data. Stores upload files (some exceeding 200 GB) via SFTP. Recently, PII was inadvertently included in these files. The company requires alerts for future PII sharing and automated remediation. What solution meets these requirements with the LEAST development effort?

- ⊗ Use an Amazon S3 bucket as a secure transfer point. Use Amazon Inspector to scan the objects in the bucket. If objects contain PII, trigger an S3 Lifecycle policy to remove the objects that contain PII.

Amazon Inspector is designed for vulnerability assessments, not PII detection. Therefore, this option is incorrect.

- ⊗ Use an Amazon S3 bucket as a secure transfer point. Use Amazon Macie to scan the objects in the bucket. If objects contain PII, use Amazon Simple Notification Service (Amazon SNS) to trigger a notification to the administrators to remove the objects that contain PII.

While Macie is designed for PII detection, it has file size limitations that preclude analyzing files exceeding 200GB. Also, this option does not fully automate remediation because it relies on the administrator to remove the object.

- ⊗ Implement custom scanning algorithms in an AWS Lambda function. Trigger the function when objects are loaded into the bucket. If objects contain PII, use Amazon Simple Notification Service (Amazon SNS) to trigger a notification to the administrators to remove the objects that contain PII.

Although this can identify PII, it does not automate remediation and requires substantial development effort. Also requires manual intervention.

- ✔ Implement custom scanning algorithms in an AWS Lambda function. Trigger the function when objects are loaded into the bucket. If objects contain PII, use Amazon Simple Email Service (Amazon SES) to trigger a notification to the administrators and trigger an S3 Lifecycle policy to remove the objects that contain PII.

This option offers automated remediation via a Lambda function and S3 Lifecycle policy. While it involves more development than using Macie, it addresses the 200GB file size constraint. However, a Lifecycle policy alone cannot selectively remove PII. The Lambda must tag the object or move it to a separate location to be cleaned up by a life cycle policy.

Overall Explanation:

Option D is the correct answer because it addresses both requirements: alerting administrators and automated remediation. While it involves more initial development effort than simply using Macie, it can handle the large file sizes (200GB+) that Macie cannot. The custom Lambda function can scan the files for PII and trigger an S3 Lifecycle policy after the file is tagged or moved which will ultimately remove the offending files.

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Question 32

Not Attempted

Security



A company needs to securely transfer data between its Salesforce SaaS application and Amazon S3. Data at rest must be encrypted using AWS KMS customer managed keys (CMKs), and data in transit must also be encrypted. API access is already enabled for the Salesforce account. Which solution meets these requirements with the least operational overhead?

- ☐ Create AWS Lambda functions to transfer the data securely from Salesforce to Amazon S3.

While Lambda can be used, it requires more manual configuration and management of encryption and data transfer logic compared to AppFlow.

- ☐ Create an AWS Step Functions workflow. Define the task to transfer the data securely from Salesforce to Amazon S3.

Step Functions, like Lambda, requires more manual configuration and management than AppFlow, increasing operational overhead.

- ☒ Create Amazon AppFlow flows to transfer the data securely from Salesforce to Amazon S3.

Amazon AppFlow is a fully managed service designed for secure data transfer between SaaS applications and AWS services. It offers built-in encryption options (at rest with KMS CMKs and in transit with SSL/TLS), simplifying the process and reducing operational overhead.

- ☐ Create a custom connector for Salesforce to transfer the data securely from Salesforce to Amazon S3.

Creating a custom connector involves significant development and maintenance effort, making it a less efficient solution compared to using AppFlow.

Overall Explanation:

Amazon AppFlow provides a managed service that supports secure data transfer between Salesforce and Amazon S3 with encryption at rest (using KMS CMKs) and in transit. It reduces operational overhead compared to solutions like Lambda, Step Functions, or custom connectors by providing pre-built integrations and simplified configuration.

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Question 33

Not Attempted

Networking



A company is developing a mobile gaming app in a single AWS Region. The app runs on multiple Amazon EC2 instances in an Auto Scaling group. The company stores the app data in Amazon DynamoDB. The app communicates by using TCP and UDP protocols between the users and the servers. The application will be used globally. The company wants to ensure the lowest possible latency for all users. Which solution will meet these requirements?

- ☐ Use AWS Global Accelerator to create an accelerator. Create an Application Load Balancer (ALB) behind an accelerator endpoint that uses Global Accelerator integration and listening on the TCP and UDP ports. Update the Auto Scaling group to register instances on the ALB.

Application Load Balancers operate at the application layer (Layer 7) and do not support UDP traffic. Global Accelerator's support for endpoints behind an ALB is not the best choice for UDP.

- ☒ Use AWS Global Accelerator to create an accelerator. Create a Network Load Balancer (NLB) behind an accelerator endpoint that uses Global Accelerator integration and listening on the TCP and UDP ports. Update the Auto Scaling group to register instances on the NLB.

This is the correct answer. AWS Global Accelerator provides static entry points for your application and routes user traffic to the nearest healthy endpoint. NLBs can handle both TCP and UDP traffic, making them suitable for the gaming app's communication needs. The Global Accelerator ensures low latency for global users, and NLB provides the scalability and high availability for your instances.

- ☐ Create an Amazon CloudFront content delivery network (CDN) endpoint. Create a Network Load Balancer (NLB) behind the endpoint and listening on the TCP and UDP ports. Update the Auto Scaling group to register instances on the NLB. Update CloudFront to use the NLB as the origin.

Amazon CloudFront is primarily designed for caching static and dynamic content, such as images, videos, and web pages. While CloudFront supports dynamic content delivery, it's not the ideal solution for real-time, low-latency communication required by a mobile gaming application that communicates via TCP and UDP. Additionally, NLB cannot be used as origin with CloudFront.

- ⊗ Create an Amazon CloudFront content delivery network (CDN) endpoint. Create an Application Load Balancer (ALB) behind the endpoint and listening on the TCP and UDP ports. Update the Auto Scaling group to register instances on the ALB. Update CloudFront to use the ALB as the origin.

Similar to Option C, CloudFront is not the optimal solution for low-latency UDP traffic used in the gaming app. Furthermore, an ALB would not be able to handle UDP traffic. Also, NLB cannot be used as origin with CloudFront.

Overall Explanation:

The correct answer is to use AWS Global Accelerator in conjunction with a Network Load Balancer (NLB). Global Accelerator provides globally distributed entry points that route traffic to the nearest healthy endpoint (NLB in this case), thus minimizing latency. The NLB can handle both TCP and UDP traffic, and the Auto Scaling group registers instances with the NLB to ensure availability.

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Question 34

Not Attempted

Databases



A company hosts an application on an Amazon EC2 instance that processes customer orders and saves them to an Amazon Aurora database. During periods of high traffic, the application struggles to process orders quickly enough. What should a solutions architect implement to ensure orders are reliably written to the database with minimal delay during peak traffic?

- ⊗ Increase the instance size of the EC2 instance when traffic is high. Write orders to Amazon Simple Notification Service (Amazon SNS). Subscribe the database endpoint to the SNS topic.

Increasing the EC2 instance size can provide temporary relief but doesn't address the underlying architecture. SNS is not suitable for reliable order processing due to its publish-subscribe nature and potential for message loss. Directly subscribing the database to SNS is not a standard practice and can cause issues.

- ✔ Write orders to an Amazon Simple Queue Service (Amazon SQS) queue. Use EC2 instances in an Auto Scaling group behind an Application Load Balancer to read from the SQS queue and process orders into the database.

This solution decouples the order processing from the application using SQS, ensuring reliable storage of orders regardless of processing capacity. An Auto Scaling group behind an ALB can dynamically scale the processing capacity based on the queue's workload, providing both reliability and performance.

- ✘ Write orders to Amazon Simple Notification Service (Amazon SNS). Subscribe the database endpoint to the SNS topic. Use EC2 instances in an Auto Scaling group behind an Application Load Balancer to read from the SNS topic.

SNS is not designed for guaranteed delivery of messages like SQS, making it unsuitable for reliable order processing. While the Auto Scaling group provides scalability, the underlying message delivery is not reliable.

- ✘ Write orders to an Amazon Simple Queue Service (Amazon SQS) queue when the EC2 instance reaches CPU threshold limits. Use scheduled scaling of EC2 instances in an Auto Scaling group behind an Application Load Balancer to read from the SQS queue and process orders into the database.

Writing to SQS only when the EC2 instance is overloaded doesn't prevent message loss during the initial overload. Scheduled scaling may not accurately respond to unpredictable traffic spikes and is less reactive than scaling based on queue depth.

Overall Explanation:

The best solution is to decouple the order processing from the initial application using SQS. This ensures that orders are reliably stored even during traffic spikes. Using an Auto Scaling group of EC2 instances behind an Application Load Balancer allows for dynamic scaling of the order processing capacity based on the number of messages in the SQS queue, providing both reliability and scalability.

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Question 35

Not Attempted

Serverless



An IoT company is launching a new smart mattress that uses sensors to track sleep data. Each mattress generates approximately 2 MB of data nightly, which is stored in an Amazon S3 bucket. The company needs to process and summarize this data for each mattress, making the

results available as quickly as possible. Data processing requires 1 GB of memory and completes within 30 seconds. What is the MOST cost-effective solution for meeting these requirements?

☐ Use AWS Glue with a Scala job

AWS Glue is a fully managed ETL service. While suitable for data transformation, it is more cost-effective for larger, more complex datasets and long-running jobs. For small, frequent tasks like this, Lambda is more efficient.

☐ Use Amazon EMR with an Apache Spark script

Amazon EMR is designed for processing large-scale data using frameworks like Spark. It involves provisioning and managing a cluster, which is overkill for the relatively small data volumes and processing time required here. This option is not cost-effective for this specific workload.

☒ Use AWS Lambda with a Python script

AWS Lambda allows you to run code without provisioning or managing servers. It is cost-effective because you only pay for the compute time consumed. Given the small data size (2 MB) and quick processing time (30 seconds), Lambda is an ideal choice. It can be triggered directly by the S3 bucket upon data arrival, ensuring low latency.

☐ Use AWS Glue with a PySpark job

Similar to the Scala job, using PySpark with AWS Glue is more suitable for larger datasets and more complex ETL pipelines. Lambda offers a more cost-effective and simpler solution for this smaller-scale data processing task.

Overall Explanation:

AWS Lambda is the most cost-effective solution because it allows for event-driven, serverless execution, perfectly suited for small data processing tasks with low latency requirements. It only charges for the compute time used, avoiding the overhead of provisioning and managing infrastructure, which would be necessary with Glue or EMR. The data processing is small, only 2 MB, and the processing requirements are 1 GB of memory for 30 seconds, all of which can easily be handled by AWS Lambda triggered by the S3 bucket data arrival.

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A company hosts an online shopping application that stores all orders in an Amazon RDS for PostgreSQL Single-AZ DB instance. Management wants to eliminate single points of failure and has asked a solutions architect to recommend an approach to minimize database downtime without requiring any changes to the application code. Which solution meets these requirements?

- ☒ Convert the existing database instance to a Multi-AZ deployment by modifying the database instance and specifying the Multi-AZ option.

Converting to a Multi-AZ deployment minimizes downtime and eliminates a single point of failure. It is simpler than creating a new instance and restoring from a snapshot, and requires no changes to the application code.

- ☐ Create a new RDS Multi-AZ deployment. Take a snapshot of the current RDS instance and restore the new Multi-AZ deployment with the snapshot.

Creating a new RDS instance and restoring from a snapshot will cause downtime during the restore process. Converting the existing instance is a more efficient approach.

- ☐ Create a read-only replica of the PostgreSQL database in another Availability Zone. Use Amazon Route 53 weighted record sets to distribute requests across the databases.

Read replicas are used for offloading read traffic and do not automatically failover in case of primary database failure, which doesn't fully address the requirement to minimize downtime without application changes.

- ☐ Place the RDS for PostgreSQL database in an Amazon EC2 Auto Scaling group with a minimum group size of two. Use Amazon Route 53 weighted record sets to distribute requests across instances.

RDS instances cannot be placed inside EC2 Auto Scaling groups. This solution is not applicable for RDS databases.

Overall Explanation:

The best approach is to convert the existing database to a Multi-AZ deployment. This eliminates the single point of failure and minimizes downtime without requiring any changes to the application code. Options involving creating new instances or manual replication introduce unnecessary complexity and potential downtime. RDS instances cannot be managed using EC2 Auto Scaling groups.

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Question 37

Not Attempted

Storage



A company is developing an application to support customer demands. The company wants to deploy the application on multiple Amazon EC2 Nitro-based instances within the same Availability Zone. The company also wants to give the application the ability to write to multiple block storage volumes in multiple EC2 Nitro-based instances simultaneously to achieve higher application availability. Which solution will meet these requirements?

- ☐ Use General Purpose SSD (gp3) EBS volumes with Amazon Elastic Block Store (Amazon EBS) Multi-Attach

gp3 volumes support Multi-Attach but may not offer the performance needed for simultaneous writes from multiple instances.

- ☐ Use Throughput Optimized HDD (st1) EBS volumes with Amazon Elastic Block Store (Amazon EBS) Multi-Attach

st1 volumes are designed for sequential workloads and are not suitable for random write-intensive operations required for simultaneous writes.

- ☒ Use Provisioned IOPS SSD (io2) EBS volumes with Amazon Elastic Block Store (Amazon EBS) Multi-Attach

io2 volumes are designed for high-performance workloads and support Multi-Attach, allowing multiple instances to simultaneously write to the same volume.

- ☐ Use General Purpose SSD (gp2) EBS volumes with Amazon Elastic Block Store (Amazon EBS) Multi-Attach

gp2 volumes support Multi-Attach but may not offer the performance needed for simultaneous writes from multiple instances.

Overall Explanation:

The correct answer is C. Amazon EBS Multi-Attach enables you to attach a single Provisioned IOPS SSD (io1 or io2) volume to multiple instances that are in the same Availability Zone. Multi-Attach is supported exclusively on Provisioned IOPS SSD (io1 and io2) volumes. Using Provisioned IOPS SSD (io2) EBS volumes with Multi-Attach provides higher performance and lower latency compared to General Purpose SSD (gp2) volumes, making io2 volumes better suited for demanding and mission-critical applications where performance is crucial.

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Question 38

Not Attempted

[Compute](#)

A company has designed a stateless two-tier application that currently uses Amazon EC2 instances in a single Availability Zone and an Amazon RDS Multi-AZ DB instance. The new management team requires that the application be highly available across multiple Availability Zones. What should a solutions architect do to meet this requirement?

- ☒ Configure the application to use Multi-AZ EC2 Auto Scaling and create an Application Load Balancer.

This option ensures high availability by distributing EC2 instances across multiple Availability Zones and using an Application Load Balancer to distribute traffic among them.

- ☐ Configure the application to take snapshots of the EC2 instances and send them to a different AWS Region.

While snapshots provide backup and DR capabilities, they don't provide immediate high availability in case of an AZ failure.

- ☐ Configure the application to use Amazon Route 53 latency-based routing to feed requests to the application.

Latency-based routing helps route users to the lowest latency endpoint, but it doesn't inherently provide high availability across multiple Availability Zones.

- ☐ Configure Amazon Route 53 rules to handle incoming requests and create a Multi-AZ Application Load Balancer.

While a Multi-AZ Application Load Balancer is good, Route 53 by itself doesn't ensure High Availability across multiple Availability Zones, as it still requires the EC2 instances to be highly available.

Overall Explanation:

To achieve high availability for the stateless two-tier application, the solutions architect should configure the application to use Multi-AZ EC2 Auto Scaling and create an Application Load Balancer. This setup will distribute EC2 instances across multiple Availability Zones, allowing the application to remain available even if one Availability Zone experiences an outage. The Application Load Balancer will distribute incoming traffic across

the healthy EC2 instances. The RDS Multi-AZ DB instance is already configured for high availability at the database tier.

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Question 39

Not Attempted

[Compute](#)

A company uses AWS Organizations. A member account has purchased a Compute Savings Plan. Due to changes in workloads, the account now utilizes less than 50% of its purchased compute power. How can the company best leverage the existing Compute Savings Plan to maximize cost savings across the organization?

- ☐ Turn on discount sharing from the Billing Preferences section of the account console in the member account that purchased the Compute Savings Plan.
This option is incorrect. Discount sharing must be enabled from the management account, not the member account.
- ☒ Turn on discount sharing from the Billing Preferences section of the account console in the company's Organizations management account.
This is the correct answer. By enabling discount sharing in the management account, the benefits of the Compute Savings Plan can be applied to other accounts within the organization.
- ☐ Migrate additional compute workloads from another AWS account to the account that has the Compute Savings Plan.
While this is a possible solution, it may involve significant effort and may not be feasible due to application dependencies or security constraints. It's not the best immediate solution.
- ☐ Sell the excess Savings Plan commitment in the Reserved Instance Marketplace.
It is not possible to sell Compute Savings Plans in the Reserved Instance Marketplace. Only Reserved Instances can be sold.

Overall Explanation:

The most efficient way to utilize an underutilized Compute Savings Plan within an AWS Organization is to enable discount sharing from the organization's management account. This allows other member accounts to benefit from the unused compute capacity. Selling

the savings plan is not an option. Migrating workloads might be possible, but potentially complex and may not be needed with discount sharing.

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Question 40

Not Attempted

Networking



A company is developing a microservices application that will provide a search catalog for customers. The application must expose REST APIs to the frontend for user interaction. The REST APIs need to access backend services that are hosted in containers within private VPC subnets. Which solution will satisfy these requirements?

- ☐ Design a WebSocket API by using Amazon API Gateway. Host the application in Amazon Elastic Container Service (Amazon ECS) in a private subnet. Create a private VPC link for API Gateway to access Amazon ECS.
This option is incorrect because the requirement specifies REST APIs, while this solution uses WebSocket APIs.
- ☒ Design a REST API by using Amazon API Gateway. Host the application in Amazon Elastic Container Service (Amazon ECS) in a private subnet. Create a private VPC link for API Gateway to access Amazon ECS.
This option is correct. It uses REST APIs with Amazon API Gateway, hosts the application in ECS within a private subnet for security, and establishes a private VPC link for secure communication between the API Gateway and the ECS containers without exposing them to the public internet.
- ☐ Design a WebSocket API by using Amazon API Gateway. Host the application in Amazon Elastic Container Service (Amazon ECS) in a private subnet. Create a security group for API Gateway to access Amazon ECS.
This option is incorrect because the requirement specifies REST APIs, while this solution uses WebSocket APIs. Additionally, a security group alone is insufficient for providing secure private connectivity; a VPC link is needed.
- ☐ Design a REST API by using Amazon API Gateway. Host the application in Amazon Elastic Container Service (Amazon ECS) in a private subnet. Create a security group for API Gateway to access Amazon ECS.
This option is incorrect. While it correctly uses REST APIs and ECS in a private subnet, a security group alone doesn't provide the necessary private and secure network connection between API Gateway and ECS. A private VPC link is the appropriate solution for this scenario.

Overall Explanation:

The correct solution involves using REST APIs with Amazon API Gateway, hosting the application in Amazon ECS within a private subnet, and creating a private VPC link for secure and private communication between the API Gateway and the ECS containers. Using a VPC link is crucial for avoiding public internet exposure and ensuring data security. The question explicitly requires REST APIs, eliminating options that suggest WebSocket APIs. Security Groups are not sufficient to secure API Gateway connections to services hosted in private VPC subnets; a VPC Link is required for that.

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Question 41

Not Attempted

Storage



A company stores raw data in an Amazon S3 bucket. This data is used for various analytics purposes on behalf of the company's customers. The analytics type determines the access pattern on the S3 objects. The company cannot predict or control the access pattern. To reduce S3 costs, which solution should the company implement?

- ☐ Use S3 replication to transition infrequently accessed objects to S3 Standard-Infrequent Access (S3 Standard-IA).

S3 Replication is used for replicating data between buckets, not for cost optimization based on access patterns.

- ☐ Use S3 Lifecycle rules to transition objects from S3 Standard to Standard-Infrequent Access (S3 Standard-IA).

While S3 Lifecycle rules can transition to S3 Standard-IA, this is suitable for known infrequent access patterns. This option is not optimal as the access pattern is unpredictable.

- ☒ Use S3 Lifecycle rules to transition objects from S3 Standard to S3 Intelligent-Tiering.

S3 Intelligent-Tiering automatically moves objects between frequent, infrequent, and archive access tiers based on changing access patterns, making it ideal for unpredictable access.

- ☐ Use S3 Inventory to identify and transition objects that have not been accessed from S3 Standard to S3 Intelligent-Tiering.

S3 Inventory helps identify objects but doesn't directly transition them. S3 Lifecycle rules are needed for the transition. While this option hints at the right direction, Intelligent-Tiering is better for unpredictable access.

Overall Explanation:

S3 Intelligent-Tiering is designed for scenarios with unpredictable access patterns. It automatically moves data between different access tiers based on how frequently it's accessed, optimizing costs without requiring manual intervention or predictions about access patterns. S3 Lifecycle rules can be used to initiate this transition from S3 Standard to S3 Intelligent-Tiering.

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Question 42

Not Attempted

[Compute](#)

A company is planning an online marketing event that will last one week and requires guaranteed Amazon EC2 capacity in three specific Availability Zones within a particular AWS Region. What is the MOST cost-effective and reliable approach to ensure they have the necessary EC2 capacity for the event?

- ☐ **Purchase Reserved Instances that specify the Region needed.**
Reserved Instances offer cost savings for long-term commitments (1 or 3 years), which is not ideal for a short-term event. They also do not guarantee capacity in specific Availability Zones.
- ☐ **Create an On-Demand Capacity Reservation that specifies the Region needed.**
This option only reserves capacity at the Region level and doesn't guarantee capacity within the three specific Availability Zones required.
- ☐ **Purchase Reserved Instances that specify the Region and three Availability Zones needed.**
Reserved Instances are not ideal for short-term capacity guarantees and do not provide capacity reservations.
- ☒ **Create an On-Demand Capacity Reservation that specifies the Region and three Availability Zones needed.**
On-Demand Capacity Reservations allow you to reserve EC2 capacity in specific Availability Zones for any duration, guaranteeing access to resources. This is the most appropriate choice for the scenario.

Overall Explanation:

For a short-term event requiring guaranteed EC2 capacity in specific Availability Zones, the most effective solution is to create an On-Demand Capacity Reservation specifying the Region and the three Availability Zones. Reserved Instances involve long-term commitments, making them unsuitable for this scenario. Options A, B and C do not guarantee the AZ requirement. Therefore Option D is correct.

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Question 43

Not Attempted

Networking



A company hosts applications on Amazon EC2 instances with IPv6 addresses. These applications need to initiate communication with external applications over the internet. The company's security policy strictly prohibits external services from initiating connections to the EC2 instances. Which solution should a solutions architect recommend to meet these requirements?

- ☐ Create a NAT gateway and make it the destination of the subnet's route table.
NAT gateways are primarily for IPv4, not IPv6. While they allow outbound traffic, they don't natively address the specific needs of IPv6 and blocking inbound connections.
- ☐ Create an internet gateway and make it the destination of the subnet's route table.
An internet gateway enables bidirectional communication. This would allow external services to initiate connections to the EC2 instances, violating the company's security policy.
- ☐ Create a virtual private gateway and make it the destination of the subnet's route table.
Virtual Private Gateways (VGWs) are used for connecting to VPNs and Direct Connect. They are not used for direct internet access from within a VPC and don't address the IPv6 or inbound connection restrictions.
- ☒ Create an egress-only internet gateway and make it the destination of the subnet's route table.
An egress-only internet gateway (EIGW) is designed specifically for IPv6 VPCs. It allows outbound IPv6 internet access while blocking all inbound IPv6 traffic, satisfying both the connectivity and security requirements.

Overall Explanation:

The correct answer is to create an egress-only internet gateway (EIGW). EIGWs are specifically designed for IPv6 and only allow outbound traffic, preventing external services from initiating connections to the EC2 instances while still allowing the EC2 instances to communicate with external services.

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Question 44

Not Attempted

Storage



A company is developing an application that runs on containers within a VPC. The application stores and accesses 1 TB of data daily in an Amazon S3 bucket. To minimize costs and ensure that traffic does not traverse the public internet, which solution should the company implement?

- ☐ Enable S3 Intelligent-Tiering for the S3 bucket
S3 Intelligent-Tiering optimizes storage costs based on access patterns but does not prevent traffic from traversing the internet.
- ☐ Enable S3 Transfer Acceleration for the S3 bucket
S3 Transfer Acceleration uses CloudFront's edge locations to accelerate uploads over the internet, which contradicts the requirement to avoid internet traffic.
- ☒ Create a gateway VPC endpoint for Amazon S3. Associate this endpoint with all route tables in the VPC
A gateway VPC endpoint allows private connectivity between the VPC and S3, ensuring traffic stays within the AWS network and minimizing data transfer costs.
- ☐ Create an interface endpoint for Amazon S3 in the VPC. Associate this endpoint with all route tables in the VPC
While an interface endpoint provides private connectivity, it is more expensive and typically used when access is needed from on-premises networks or peered VPCs.

Overall Explanation:

Creating a gateway VPC endpoint for Amazon S3 and associating it with all route tables in the VPC ensures that traffic between the VPC and S3 remains within the AWS network, minimizing internet traffic and associated costs. S3 Intelligent-Tiering addresses storage costs based on access patterns, while S3 Transfer Acceleration accelerates transfers over

the internet. Interface endpoints, while providing private connectivity, are more expensive and are not necessary in this scenario, as the application resides within the VPC.

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Question 45

Not Attempted

Databases



A company has a mobile chat application using Amazon DynamoDB as its data store. Users need to access new messages with the lowest possible latency. A solutions architect must design an optimal solution that requires minimal changes to the existing application code. Which approach is the MOST suitable?

- ☒ Configure Amazon DynamoDB Accelerator (DAX) for the new messages table. Update the code to use the DAX endpoint.

DAX is an in-memory cache specifically designed for DynamoDB. It provides low-latency access and requires only a minimal code change to point to the DAX endpoint. This option minimizes application changes while significantly reducing read latency.

- ☐ Add DynamoDB read replicas to handle the increased read load. Update the application to point to the read endpoint for the read replicas.

While read replicas can improve read throughput, they don't guarantee the lowest latency due to replication lag. Additionally, directing reads to replicas still requires application changes.

- ☐ Double the number of read capacity units for the new messages table in DynamoDB. Continue to use the existing DynamoDB endpoint.

Increasing read capacity can improve throughput but does not address latency as effectively as an in-memory cache. It also incurs higher costs without necessarily providing the required latency improvements.

- ☐ Add an Amazon ElastiCache for Redis cache to the application stack. Update the application to point to the Redis cache endpoint instead of DynamoDB.

Using ElastiCache for Redis can provide low latency. However, integrating a new caching layer would involve more significant changes to the application code than using DAX, making it less aligned with the requirement of minimal application changes.

Overall Explanation:

DynamoDB Accelerator (DAX) is the optimal choice because it provides in-memory caching for DynamoDB, minimizing latency for reading new messages. Importantly, it requires only

minimal changes to the application code to switch to the DAX endpoint, satisfying the design constraint. Read replicas improve throughput but not necessarily latency. Increasing read capacity is costly and less effective. ElastiCache requires extensive code changes.

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Question 46

Not Attempted

Networking



A company hosts a website serving static content on Amazon EC2 instances behind an Application Load Balancer (ALB). Website traffic is increasing, and the company is concerned about potential cost increases. What is the MOST cost-effective solution to reduce the load on the EC2 instances and minimize data transfer costs?

- ☒ Create an Amazon CloudFront distribution to cache static files at edge locations.
CloudFront is a content delivery network (CDN) service that caches content at edge locations worldwide. By creating a CloudFront distribution, static content from the website can be cached at edge locations, reducing the load on the EC2 instances and improving the overall performance and lowering data transfer costs.
- ☐ Create an Amazon ElastiCache cluster. Connect the ALB to the ElastiCache cluster to serve cached files.
ElastiCache is primarily used for caching dynamic content or frequently accessed data within the application tier. While it can improve performance, it's not as cost-effective as CloudFront for serving static content.
- ☐ Create an AWS WAF web ACL and associate it with the ALB. Add a rule to the web ACL to cache static files.
AWS WAF is a web application firewall and primarily protects against web exploits. It does not provide caching capabilities for static content.
- ☐ Create a second ALB in an alternative AWS Region. Route user traffic to the closest Region to minimize data transfer costs.
Deploying a second ALB in another region will increase costs and complexity. While it might reduce latency for some users, it's not the most cost-effective solution for caching static content.

Overall Explanation:

Creating an Amazon CloudFront distribution is the most cost-effective solution. CloudFront caches static content at edge locations closer to the users. This reduces the load on the

EC2 instances, decreases data transfer costs, and improves website performance by delivering content with lower latency.

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Question 47

Not Attempted

Networking



A company operates multiple VPCs across various AWS Regions to ensure isolation of workloads. Due to a new application requirement, all VPCs across all Regions must now communicate with each other. Which solution will satisfy this requirement with the LEAST amount of administrative overhead?

- ☐ Use VPC peering to manage VPC communication in a single Region. Use VPC peering across Regions to manage VPC communications.
VPC peering can become complex and difficult to manage across numerous VPCs and Regions, requiring individual peering connections between each VPC pair.
- ☐ Use AWS Direct Connect gateways across all Regions to connect VPCs across regions and manage communications.
Direct Connect is primarily for connecting on-premises networks to AWS and is not the ideal solution for VPC-to-VPC communication within AWS.
- ☒ Use AWS Transit Gateway to manage VPC communication in a single Region and Transit Gateway peering across Regions to manage VPC communications.
Transit Gateway simplifies network connectivity between VPCs and on-premises networks. Transit Gateway peering allows connecting Transit Gateways in different regions, creating a scalable and manageable solution.
- ☐ Use AWS PrivateLink across all Regions to connect VPCs across Regions and manage VPC communications.
PrivateLink is designed to provide private connectivity between VPCs and supported AWS services, services hosted by other AWS accounts (referred to as SaaS vendors), and supported AWS Marketplace partner services. It is not the correct solution for general VPC-to-VPC communication across all regions.

Overall Explanation:

AWS Transit Gateway provides a centralized hub for managing VPC communication within a single region and Transit Gateway peering enables connectivity across regions. This

solution minimizes administrative overhead compared to managing numerous individual VPC peering connections. Direct Connect is for on-premise connectivity and PrivateLink provides secure access to services, not general VPC-to-VPC communication.

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Question 48

Not Attempted

Storage



A company is designing a containerized application that will use Amazon Elastic Container Service (Amazon ECS). The application needs to access a shared file system that is highly durable and can recover data to another AWS Region with a recovery point objective (RPO) of 8 hours. The file system needs to provide a mount target in each Availability Zone within a Region. A solutions architect wants to use AWS Backup to manage the replication to another Region. Which solution will meet these requirements?

☐ Amazon FSx for Windows File Server with a Multi-AZ deployment

Amazon FSx for Windows File Server is primarily for Windows-based applications and is not the best fit for general containerized applications on ECS. Also, while it offers high availability, AWS Backup integration for cross-region replication might not be as straightforward for meeting the specific RPO.

☐ Amazon FSx for NetApp ONTAP with a Multi-AZ deployment

While FSx for NetApp ONTAP is a powerful and versatile file system, it's often an over-engineered and more expensive solution for simple shared file storage with cross-region replication. Though possible, using NetApp ONTAP just for backup storage is not cost-effective. Also, it doesn't natively support replication to another region through AWS Backup.

☒ Amazon Elastic File System (Amazon EFS) with the Standard storage class

Amazon EFS is a fully managed, scalable, and elastic file system designed for use with AWS Cloud services and on-premises resources. It provides a mount target in each Availability Zone within a Region and integrates well with ECS. Furthermore, AWS Backup can be used to manage replication of EFS to another region. The 8-hour RPO can be met with AWS Backup.

☐ Amazon FSx for OpenZFS

Amazon FSx for OpenZFS requires a Linux kind of host for access and is not a built-in file system in AWS by default, and not the best choice for general ECS use cases where a simple, shared, and scalable file system is needed.

Overall Explanation:

Amazon EFS with the Standard storage class is the most appropriate solution. It natively supports mount targets in each AZ, integrates well with ECS, and AWS Backup can manage replication to another region to meet the 8-hour RPO. Options A, B and D are either overly complex or not designed for this particular use case. FSx for NetApp ONTAP can replicate but EFS is more cost effective for this particular use case.

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Question 49

Not Attempted

Security



A company anticipating rapid growth needs to configure existing users and grant permissions to new users on AWS. The solutions architect plans to use IAM groups, adding new users to groups based on their department. Which action represents the MOST secure approach for granting permissions to these new users?

- ☐ ☒ Apply service control policies (SCPs) to manage access permissions.
SCPs are used to manage permissions across AWS Organizations and do not directly grant permissions to IAM users or groups. They set boundaries rather than providing access.
- ☐ ☒ Create IAM roles that have least privilege permission. Attach the roles to the IAM groups.
IAM roles are not directly attached to IAM groups. Roles are assumed by users, services, or applications.
- ☒ ☐ Create an IAM policy that grants least privilege permission. Attach the policy to the IAM groups.
This is the most secure approach. IAM policies define permissions and can be attached to IAM groups, allowing you to manage permissions centrally and grant only the necessary privileges.
- ☐ ☒ Create IAM roles. Associate the roles with a permissions boundary that defines the maximum permissions.
While permissions boundaries limit the maximum permissions a role can have, this option doesn't directly address granting permissions to users within IAM groups in the most efficient and secure manner.

Overall Explanation:

The most secure way to grant permissions to new users in IAM groups is to create an IAM policy that grants the least privilege necessary and attach it to the IAM groups. This allows for centralized management of permissions and ensures users only have the access they need. SCPs are for setting organizational guardrails, and IAM roles are not directly attached to IAM groups. Permissions boundaries limit maximum permissions but don't define the specific permissions granted.

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Question 50

Not Attempted

Security



A group requires permissions to list an Amazon S3 bucket and delete objects from that bucket. An administrator has created the following IAM policy to provide access to the bucket and applied that policy to the group. The group is not able to delete objects in the bucket. The company follows least-privilege access rules.

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Action": [
        "s3:ListBucket",
        "s3:DeleteObject"
      ],
      "Resource": [
        "arn:aws:s3:::bucket-name"
      ],
      "Effect": "Allow"
    }
  ]
}
```

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Which statement should a solutions architect add to the policy to correct bucket access?

- ☒ json { "Effect": "Allow", "Action": "s3:ListBucket", "Resource":
"arn:aws:s3:::examplebucket" }

This option only allows listing the bucket, but it doesn't address the need to delete objects.

- ⊗ `json { "Effect": "Allow", "Action": "s3:*", "Resource": "arn:aws:s3:::examplebucket/*" }`

This option grants all S3 permissions, violating the least-privilege principle. It also applies to objects only, not the bucket itself.

- ⊗ `json { "Effect": "Allow", "Action": "s3:*", "Resource": "arn:aws:s3:::examplebucket" }`

This option grants all S3 permissions on the bucket, violating the least-privilege principle. It does not allow access to the bucket contents.

- ✓ `json { "Effect": "Allow", "Action": "s3:DeleteObject", "Resource": "arn:aws:s3:::examplebucket/*" }`

This option provides the specific permission required to delete objects within the S3 bucket and adheres to the least-privilege principle. It targets objects within the bucket.

Overall Explanation:

The policy requires the ability to delete objects. The correct approach is to grant only the `s3:DeleteObject` permission specifically to objects within the bucket (specified by `/\*`). This adheres to the principle of least privilege.

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Question 51

Not Attempted

Security



A law firm needs to share information with the public. The information includes hundreds of files that must be publicly readable. Modifications or deletions of the files by anyone before a designated future date are prohibited. Which solution will meet these requirements in the MOST secure way?

- ⊗ Upload all files to an Amazon S3 bucket that is configured for static website hosting. Grant read-only IAM permissions to any AWS principals that access the S3 bucket until the designated date.

This option does not prevent modifications or deletions. Read-only IAM permissions do not provide object-level immutability.

- ✓ Create a new Amazon S3 bucket with S3 Versioning enabled. Use S3 Object Lock with a retention period in accordance with the designated date. Configure the S3 bucket for

static website hosting. Set an S3 bucket policy to allow read-only access to the objects. This is the most secure option. S3 Versioning allows for recovery if an object is accidentally deleted. S3 Object Lock prevents objects from being deleted or overwritten for a specified retention period. A bucket policy grants read-only access to the public.

- ⊗ Create a new Amazon S3 bucket with S3 Versioning enabled. Configure an event trigger to run an AWS Lambda function in case of object modification or deletion. Configure the Lambda function to replace the objects with the original versions from a private S3 bucket.

This option is more complex and less secure than using S3 Object Lock. It relies on an event trigger and a Lambda function to restore objects, which introduces potential points of failure.

- ⊗ Upload all files to an Amazon S3 bucket that is configured for static website hosting. Select the folder that contains the files. Use S3 Object Lock with a retention period in accordance with the designated date. Grant read-only IAM permissions to any AWS principals that access the S3 bucket.

S3 manages objects, not folders. Also, applying Object Lock on the folder level is not possible. Read-only IAM permissions do not prevent object-level changes before the designated date; Object Lock ensures the immutability.

Overall Explanation:

Option B is the most secure because it utilizes S3 Versioning and S3 Object Lock. S3 Versioning protects against accidental deletion, and S3 Object Lock ensures that objects cannot be modified or deleted for a specific retention period. S3 bucket policy ensures read only access for the objects. Options A and D do not have object-level immutability features. Option C is overly complex and relies on a Lambda function for restoration, which introduces more potential points of failure. Object Lock works only in buckets that have S3 Versioning enabled.

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Question 52

Not Attempted

Automation



A company is making a prototype of the infrastructure for its new website by manually provisioning the necessary infrastructure. This infrastructure includes an Auto Scaling group, an Application Load Balancer, and an Amazon RDS database. After the configuration has been thoroughly validated, the company wants the capability to immediately deploy the

infrastructure for development and production use in two Availability Zones in an automated fashion. What should a solutions architect recommend to meet these requirements?

- ⊗ Use AWS Systems Manager to replicate and provision the prototype infrastructure in two Availability Zones.

AWS Systems Manager is not primarily designed for infrastructure as code and automated deployments in the way described. It focuses more on managing existing instances.

- ✓ Define the infrastructure as a template by using the prototype infrastructure as a guide. Deploy the infrastructure with AWS CloudFormation.

AWS CloudFormation is the correct choice because it allows you to define and provision infrastructure as code. You can create a template representing the infrastructure and deploy it automatically across multiple Availability Zones.

- ⊗ Use AWS Config to record the inventory of resources that are used in the prototype infrastructure. Use AWS Config to deploy the prototype infrastructure into two Availability Zones.

AWS Config is primarily for configuration management and compliance, not for initial infrastructure provisioning and deployment.

- ⊗ Use AWS Elastic Beanstalk and configure it to use an automated reference to the prototype infrastructure to automatically deploy new environments in two Availability Zones.

AWS Elastic Beanstalk is better suited for deploying and managing applications, rather than defining the underlying infrastructure (Auto Scaling Group, Load Balancer, RDS). While it can handle EC2 deployments well, the RDS component is better handled by Infrastructure as Code like CloudFormation.

Overall Explanation:

The best solution is to use AWS CloudFormation. CloudFormation allows you to define the entire infrastructure (Auto Scaling group, Application Load Balancer, and RDS database) as a template. This template can then be used to automatically deploy the infrastructure in multiple Availability Zones, ensuring consistency and repeatability. This approach embodies the principle of Infrastructure as Code. Elastic Beanstalk is more geared toward application deployment and management and isn't the ideal tool for provisioning the full infrastructure stack in this scenario. Systems Manager and Config are not designed for this type of infrastructure-as-code deployment.

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Question 53

Not Attempted

Storage



A company's website relies on an Amazon EC2 instance store for hosting its product catalog. To ensure high availability and data durability for the catalog, what action should a solutions architect take?

☐ Move the catalog to Amazon ElastiCache for Redis.

ElastiCache is an in-memory data store primarily used for caching, which doesn't provide durable storage. While it can be configured for persistence, it's not designed as a primary durable storage solution.

☐ Deploy a larger EC2 instance with a larger instance store.

Increasing the size of the instance store doesn't address the durability or availability requirements. Instance stores are ephemeral, meaning data is lost when the instance is stopped, terminated, or fails.

☐ Move the catalog from the instance store to Amazon S3 Glacier Deep Archive.

Glacier Deep Archive is designed for long-term archival storage and is not suitable for frequently accessed data due to long retrieval times.

☒ Move the catalog to an Amazon Elastic File System (Amazon EFS) file system.

Amazon EFS provides a scalable, fully managed file system accessible to multiple EC2 instances concurrently. It offers high availability and durability through data replication across Availability Zones.

Overall Explanation:

The instance store lacks durability. ElastiCache is memory-based. Glacier is for archival. EFS provides a highly available and durable network file system suitable for this use case. Therefore, moving the catalog to Amazon EFS is the correct solution.

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Question 54

Not Attempted

Networking



A business application is hosted on Amazon EC2 and uses Amazon S3 for encrypted object storage. The chief information security officer has directed that no application traffic between the two services should traverse the public internet. Which capability should the solutions architect use to meet the compliance requirements?

☐ AWS Key Management Service (AWS KMS)

AWS KMS is used for managing encryption keys and does not prevent traffic from traversing the public internet.

☒ VPC endpoint

A VPC endpoint enables you to privately access AWS services without requiring internet gateways, NAT gateways, VPN connections, or AWS Direct Connect connections. It allows you to connect your VPC directly to supported AWS services, such as Amazon S3, over a private connection within the AWS network.

☐ Private subnet

A private subnet prevents direct internet access for instances within the subnet but does not, on its own, ensure that traffic to S3 doesn't traverse the public internet.

☐ Virtual private gateway

A virtual private gateway is used to create VPN connections between your VPC and on-premises networks, and doesn't directly address the requirement of preventing S3 traffic from traversing the public internet.

Overall Explanation:

A VPC endpoint for Amazon S3 allows traffic between EC2 instances and S3 to remain within the AWS network, avoiding the public internet. This fulfills the compliance requirement of ensuring no application traffic traverses the public internet.

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Question 55

Not Attempted

Databases



A company hosts a three-tier web application in the AWS Cloud. A Multi-AZ Amazon RDS for MySQL server forms the database layer, and Amazon ElastiCache forms the cache layer. The company requires a caching strategy that adds or updates data in the cache synchronously

when a customer adds an item to the database, ensuring the cache data always matches the database data. Which solution meets these requirements?

☐ Implement the lazy loading caching strategy.

Lazy loading caches data only when it is requested, which doesn't guarantee immediate synchronization with the database after a write operation.

☒ Implement the write-through caching strategy.

The write-through caching strategy updates the cache at the same time as the database, ensuring data consistency.

☐ Implement the adding TTL caching strategy.

Adding TTL (Time To Live) only defines how long data stays in the cache, not how it gets initially populated or updated after database changes.

☐ Implement the AWS AppConfig caching strategy.

AWS AppConfig is used for configuration management, not for caching data in sync with database updates.

Overall Explanation:

The write-through caching strategy is the correct choice because it updates the cache synchronously with the database, ensuring the cache data always matches the database data. When a customer adds or updates an item in the database, the application first writes the data to the database and then updates the cache with the same data. This ensures that the cache is always synchronized with the database, as every write operation triggers an update to the cache.

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Question 56

Not Attempted

Storage



A company needs to migrate 100 GB of data from its on-premises data center to an Amazon S3 bucket. They have a 100 Mbps internet connection and require the data to be encrypted in transit. New data will be written directly to Amazon S3 moving forward. Which solution provides the LEAST operational overhead while meeting these requirements?

- ✔ Use the `s3 sync` command in the AWS CLI to transfer the data directly to the S3 bucket.

Using `s3 sync` is a straightforward approach. The AWS CLI automatically encrypts data in transit via TLS. With a 100 Mbps connection, transferring 100 GB would take approximately 2.3 hours, assuming a stable network. This option requires minimal upfront configuration.

- ✘ Use AWS DataSync to migrate the data from the on-premises location to the S3 bucket.
AWS DataSync is a fully managed data transfer service that handles encryption in transit and automates the data transfer process. However, it requires installing and configuring an agent, which adds operational overhead compared to the AWS CLI.

- ✘ Use AWS Snowball to move the data to the S3 bucket.

Using AWS Snowball would be an overkill solution for only 100GB of data, introducing unnecessary complexity and delays associated with shipping the physical device.

- ✘ Set up an IPsec VPN from the on-premises location to AWS, then use the `s3 cp` command in the AWS CLI to move the data directly to the S3 bucket.

Setting up an IPsec VPN adds significant operational overhead and is unnecessary since the AWS CLI already provides encryption in transit. Using `s3 cp` would also require scripting for larger file transfers.

Overall Explanation:

The most straightforward and efficient solution with the least operational overhead is to use the `s3 sync` command in the AWS CLI. The AWS CLI automatically encrypts data in transit using TLS, and a 100 Mbps connection is sufficient for transferring 100 GB of data. AWS DataSync and AWS Snowball introduce more complexity. Setting up an IPsec VPN is unnecessary as the AWS CLI already handles encryption.

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Question 57

Not Attempted

Compute



A company has containerized a Windows job that runs on the .NET 6 Framework within a Windows container. The company aims to execute this job within the AWS Cloud. The job is scheduled to run every 10 minutes, with a runtime that varies between 1 and 3 minutes. Which solution offers the MOST cost-effective approach to meet these requirements?

- ⊗ Create an AWS Lambda function based on the container image of the job. Configure Amazon EventBridge to invoke the function every 10 minutes.

AWS Lambda does not support Windows-based container images. Lambda only supports Linux-based container images, making this option unsuitable.

- ⊗ Use AWS Batch to create a job that uses AWS Fargate resources. Configure the job scheduling to run every 10 minutes.

AWS Batch now supports Fargate and Windows Containers, and this can be a viable option.

- ✔ Use Amazon Elastic Container Service (Amazon ECS) on AWS Fargate to run the job. Create a scheduled task based on the container image of the job to run every 10 minutes.

Using Amazon ECS on AWS Fargate enables the job to run in a containerized environment while leveraging the serverless nature of Fargate, where you are charged only for the resources consumed during the job's execution. Creating a scheduled task ensures it runs every 10 minutes as required. ECS supports Windows containers and scheduled tasks natively.

- ⊗ Use Amazon Elastic Container Service (Amazon ECS) on AWS Fargate to run the job. Create a standalone task based on the container image of the job. Use Windows task scheduler to run the job every 10 minutes.

Using Windows Task Scheduler within the container is not a scalable or manageable solution for a cloud environment. It requires the container to run continuously and is not cost-effective.

Overall Explanation:

Amazon ECS on Fargate provides a cost-effective and scalable solution for running containerized Windows jobs on a schedule. It avoids the overhead of managing servers and ensures that resources are only consumed during the job's execution. While AWS Batch is now a viable option it requires extra overhead compared to ECS alone. Lambda is not possible because it does not support Windows containers.

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Question 58

Not Attempted

Security



A company is migrating from multiple standalone AWS accounts to a consolidated, multi-account environment. They plan to establish many new AWS accounts for distinct business units. A key requirement is to authenticate user access to these AWS accounts using the

company's existing centralized corporate directory service. Which combination of actions should a solutions architect recommend to fulfill these requirements?

- ☒ Create a new organization in AWS Organizations with all features turned on. Create the new AWS accounts within this organization.

Creating an AWS Organization provides the foundational structure for managing multiple accounts centrally and enables the use of services like IAM Identity Center.

- ☐ Set up an Amazon Cognito identity pool. Configure AWS IAM Identity Center (AWS Single Sign-On) to accept Amazon Cognito authentication.

Amazon Cognito is for customer identity and access management, it is not suited for integrating with a corporate directory.

- ☐ Configure a service control policy (SCP) to manage the AWS accounts. Add AWS IAM Identity Center (AWS Single Sign-On) to AWS Directory Service.

SCPs are for governance, not authentication. Adding IAM Identity Center to AWS Directory Service isn't a standard practice.

- ☐ Create a new organization in AWS Organizations. Configure the organization's authentication mechanism to use AWS Directory Service directly.

AWS Organizations itself does not directly integrate with directory services for authentication. IAM Identity Center handles this integration.

- ☒ Set up AWS IAM Identity Center (AWS Single Sign-On) in the organization. Configure IAM Identity Center, and integrate it with the company's corporate directory service.

AWS IAM Identity Center (successor to AWS Single Sign-On) enables centralized access management across multiple AWS accounts and allows integration with corporate directory services for authentication.

Overall Explanation:

Creating an AWS Organization provides the multi-account management structure. Implementing AWS IAM Identity Center (successor to AWS Single Sign-On) and integrating it with the company's corporate directory service enables centralized authentication and access control across the AWS accounts within the Organization. Therefore, these two actions combined satisfy the requirements.

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Question 59

Not Attempted

Storage



A company is archiving old news footage in AWS and needs a cost-effective solution for storing these video files. The files will rarely be restored, but when needed, they must be available within five minutes. What is the MOST cost-effective storage solution that meets these requirements?

- ☒ Store the video archives in Amazon S3 Glacier and use Expedited retrievals.
Amazon S3 Glacier provides low-cost archival storage, and Expedited retrievals allow for data retrieval within 1-5 minutes, fulfilling the availability requirement. While Expedited retrievals are more expensive than Standard retrievals, the overall cost savings from Glacier compared to other S3 storage classes makes it a cost-effective solution when infrequent access with quick retrieval times are needed.
- ☐ Store the video archives in Amazon S3 Glacier and use Standard retrievals.
Amazon S3 Glacier provides low-cost archival storage, but standard retrievals can take several hours, not meeting the five-minute requirement.
- ☐ Store the video archives in Amazon S3 Standard-Infrequent Access (S3 Standard-IA).
S3 Standard-IA is designed for data accessed less frequently but still requires rapid access. While faster than Glacier with Standard retrieval, it is more expensive for storage than Glacier, making it less cost-effective for archiving.
- ☐ Store the video archives in Amazon S3 One Zone-Infrequent Access (S3 One Zone-IA).
S3 One Zone-IA offers lower storage costs than S3 Standard-IA but stores data in a single Availability Zone. This makes it more susceptible to data loss and does not provide significant cost savings compared to Glacier for archival purposes while being more expensive in general.

Overall Explanation:

The most cost-effective solution for archiving video footage with infrequent access and a five-minute retrieval requirement is to store the files in Amazon S3 Glacier and use Expedited retrievals. Glacier offers the lowest storage costs for archival data, and Expedited retrievals ensure the data can be accessed quickly when needed. While Expedited retrievals have a higher cost per retrieval, the overall storage cost savings outweigh the retrieval costs, making it the best option. S3 Standard-IA and S3 One Zone-IA are more expensive for storage, and Glacier Standard retrievals do not meet the time constraint.

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Question 60

Not Attempted

Automation



A company is developing a three-tier web application on AWS, aiming for simplified deployment and reduced operational costs. The presentation tier consists of a static website. The application logic is containerized. The data tier uses a relational database. Which architecture best fulfills these requirements?

- ☒ Use Amazon S3 to host static content. Use Amazon Elastic Container Service (Amazon ECS) with AWS Fargate for compute power. Use a managed Amazon RDS cluster for the database.

This solution leverages cost-effective and scalable services. S3 is ideal for static websites. ECS with Fargate eliminates the need to manage EC2 instances for containers, reducing operational overhead. RDS offloads database management tasks to AWS, ensuring high availability and durability.

- ☐ Use Amazon CloudFront to host static content. Use Amazon Elastic Container Service (Amazon ECS) with Amazon EC2 for compute power. Use a managed Amazon RDS cluster for the database.

While CloudFront can host static content, it adds unnecessary cost for a simple static website, since the CDN functionality might not be required. Using EC2 directly with ECS increases operational overhead compared to Fargate.

- ☐ Use Amazon S3 to host static content. Use Amazon Elastic Kubernetes Service (Amazon EKS) with AWS Fargate for compute power. Use a managed Amazon RDS cluster for the database.

EKS with Fargate is a viable option but is more complex to manage and potentially more expensive than ECS with Fargate for many use cases.

- ☐ Use Amazon EC2 Reserved Instances to host static content. Use Amazon Elastic Kubernetes Service (Amazon EKS) with Amazon EC2 for compute power. Use a managed Amazon RDS cluster for the database.

Using EC2 instances directly for the presentation and logic tier results in higher operational overhead and complexity. S3 is better suited for static website hosting. Although Reserved Instances can offer cost savings, it requires capacity planning and upfront commitment.

Overall Explanation:

The most suitable solution is to use Amazon S3 for static content hosting due to its cost-effectiveness and scalability. Amazon ECS with AWS Fargate is preferred for containerized applications because it eliminates infrastructure management. Utilizing a managed Amazon RDS cluster simplifies database administration, ensures high availability, and reduces operational burden. Therefore option A is correct.

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Question 61

Not Attempted

Storage



A company requires a storage solution for its application. The solution must be highly available and scalable. The storage solution needs to function as a file system, be mountable by multiple Linux instances both in AWS and on-premises using native protocols, and have no minimum size requirements. The company has established a Site-to-Site VPN connection to enable access from its on-premises network to its VPC. Which storage solution best meets these requirements?

☐ Amazon FSx Multi-AZ deployments

Amazon FSx is a managed file system service, but it does not support native protocols like NFS, which are required for mounting on Linux instances and on-premises systems via VPN.

☐ Amazon Elastic Block Store (Amazon EBS) Multi-Attach volumes

Amazon EBS is a block storage service. While EBS Multi-Attach allows attaching a volume to multiple EC2 instances, it cannot be mounted by multiple Linux instances through native protocols such as NFS, nor can it be directly accessed from on-premises environments.

☒ Amazon Elastic File System (Amazon EFS) with multiple mount targets

Amazon EFS is a fully managed file system service that provides scalable, shared storage for Amazon EC2 instances and on-premises servers. It supports the Network File System version 4 (NFSv4) protocol, which is a native protocol for Linux-based systems. Multiple mount targets allow access from multiple EC2 instances within a VPC, and it can also be accessed from on-premises environments via VPN or Direct Connect.

☐ Amazon Elastic File System (Amazon EFS) with a single mount target and multiple access points

While EFS itself is the correct storage service, using only a single mount target limits the availability and scalability of the solution. Also, access points are designed for controlling access and not for

enabling multiple independent instances to mount the file system across different AZs or from on-premises.

Overall Explanation:

Amazon EFS with multiple mount targets is the best solution. It provides a scalable and highly available file system that can be mounted by multiple Linux instances both in AWS and on-premises via NFS over a Site-to-Site VPN. EFS natively supports POSIX file system semantics. FSx does not use native protocols, EBS is a block storage and cannot be mounted on-prem, and a single mount target limits availability. Access points control access and are not a substitute for multiple mount targets.

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Question 62

Not Attempted

Security



A 4-year-old media company is using the AWS Organizations all features feature set to organize its AWS accounts. The company's finance team requires that the billing information on the member accounts must not be accessible to anyone, including the root user of the member accounts. Which solution will meet these requirements?

- ☐ Add all finance team users to an IAM group. Attach an AWS managed policy named Billing to the group.

This solution grants billing access, rather than denying it, and doesn't prevent root users from accessing billing information.

- ☐ Attach an identity-based policy to deny access to the billing information to all users, including the root user.

Identity-based policies cannot override the root user's inherent permissions.

- ☒ Create a service control policy (SCP) to deny access to the billing information. Attach the SCP to the root organizational unit (OU).

This is the correct solution. SCPs can restrict permissions, including for the root user, within member accounts of the OU to which they're attached.

- ☐ Convert from the Organizations all features feature set to the Organizations consolidated billing feature set.

Consolidated billing addresses billing aggregation and payment, but doesn't restrict access to billing data within member accounts.

Overall Explanation:

Service Control Policies (SCPs) provide central control over the maximum available permissions for all accounts in your organization. By creating an SCP and attaching it to the root OU, you can explicitly deny access to billing information for all accounts within the organization. This ensures that no member accounts, including root users, have access to the billing information. Therefore, the correct answer is C.

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Question 63

Not Attempted

Serverless



An ecommerce company runs an application in the AWS Cloud that is integrated with an on-premises warehouse solution. The company uses Amazon Simple Notification Service (Amazon SNS) to send order messages to an on-premises HTTPS endpoint so the warehouse application can process the orders. The local data center team has detected that some of the order messages were not received. A solutions architect needs to retain messages that are not delivered and analyze the messages for up to 14 days. Which solution will meet these requirements with the LEAST development effort?

- ☐ Configure an Amazon SNS dead letter queue that has an Amazon Kinesis Data Stream target with a retention period of 14 days.

Using Kinesis Data Streams as a dead letter queue target would require significant custom coding and is not the least effort solution.

- ☐ Add an Amazon Simple Queue Service (Amazon SQS) queue with a retention period of 14 days between the application and Amazon SNS.

Adding an SQS queue between the application and SNS would require modifying the application logic and would not leverage the dead letter queue functionality directly. It also wouldn't inherently prevent failed deliveries to the on-premises endpoint.

- ☒ Configure an Amazon SNS dead letter queue that has an Amazon Simple Queue Service (Amazon SQS) target with a retention period of 14 days.

This is the correct answer. SNS dead letter queues are designed to handle undeliverable messages. Using SQS as the target allows for retention and analysis of those messages with minimal

configuration. SQS retention period can be configured up to 14 days.

- ⊗ Configure an Amazon SNS dead letter queue that has an Amazon DynamoDB target with a TTL attribute set for a retention period of 14 days.

Using DynamoDB as a dead letter queue target would require custom coding and is not the least effort solution.

Overall Explanation:

The least effort solution is to utilize the built-in dead letter queue functionality of SNS. Configuring an SNS dead letter queue with an SQS target, and setting the SQS queue's retention period to 14 days, allows for retaining undelivered messages for analysis without requiring custom code. SNS dead letter queues are by default SQS queues. Other options require significant code changes or do not correctly utilize SNS's dead letter queue feature.

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Question 64

Not Attempted

[Storage](#)



A company stores call transcript files on a monthly basis. Users access the files randomly within 1 year of the call, but users access the files infrequently after 1 year. The company wants to optimize its solution by giving users the ability to query and retrieve files that are less than 1-year-old as quickly as possible. A delay in retrieving older files is acceptable. Which solution will meet these requirements MOST cost-effectively?

- ⊗ Store individual files with tags in Amazon S3 Glacier Instant Retrieval. Query the tags to retrieve the files from S3 Glacier Instant Retrieval.

Storing solely in Glacier Instant Retrieval might not be the most cost-effective as it's designed for infrequent access with immediate retrieval, not frequent random access within the first year. It also requires querying using tags which might not be the most efficient approach.

- ✔ Store individual files in Amazon S3 Intelligent-Tiering. Use S3 Lifecycle policies to move the files to S3 Glacier Flexible Retrieval after 1 year. Query and retrieve the files that are in Amazon S3 by using Amazon Athena. Query and retrieve the files that are in S3 Glacier by using S3 Glacier Select.

This option leverages S3 Intelligent-Tiering for frequent access in the first year, then transitions to Glacier Flexible Retrieval for cost savings after 1 year. Using Athena and Glacier Select for querying provides a cost-effective way to search and retrieve data in both storage classes. The delay in retrieving older files is also acceptable according to the question.

- ✘ Store individual files with tags in Amazon S3 Standard storage. Store search metadata for each archive in Amazon S3 Standard storage. Use S3 Lifecycle policies to move the files to S3 Glacier Instant Retrieval after 1 year. Query and retrieve the files by searching for metadata from Amazon S3.

Storing search metadata in S3 Standard can be costly, especially if the metadata is extensive. While using Glacier Instant Retrieval for older files meets the requirement, this is not as cost-effective as using a flexible retrieval option.

- ✘ Store individual files in Amazon S3 Standard storage. Use S3 Lifecycle policies to move the files to S3 Glacier Deep Archive after 1 year. Store search metadata in Amazon RDS. Query the files from Amazon RDS. Retrieve the files from S3 Glacier Deep Archive.

Using Amazon RDS for metadata storage adds unnecessary complexity and cost. Glacier Deep Archive has the longest retrieval times, which contradicts the requirement of quick retrieval for files less than 1 year old. Storing all the files in Standard Storage for a year also makes it expensive.

Overall Explanation:

Option B is the most cost-effective solution. It utilizes S3 Intelligent-Tiering to optimize storage costs based on access patterns, automatically moving files to the most appropriate tier within the first year to ensure quick access. Then, it transitions older files to S3 Glacier Flexible Retrieval for long-term archival and cost savings since a delay in accessing older files is acceptable. Finally, using Athena and Glacier Select is efficient for querying data in S3 and Glacier respectively. Other options are either more expensive (using only S3 Standard or Glacier Instant Retrieval) or more complex (using RDS for metadata).

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Question 65

Not Attempted

Databases



A gaming company uses Amazon DynamoDB to store user information such as geographic location, player data, and leaderboards. The company needs to configure continuous backups to an Amazon S3 bucket with a minimal amount of coding. The backups must not affect the availability of the application and must not affect the read capacity units (RCUs) that are defined for the table. Which solution meets these requirements?

- ☐ Use an Amazon EMR cluster. Create an Apache Hive job to back up the data to Amazon S3.

This solution requires significant coding and server management, and it doesn't directly use DynamoDB's built-in backup capabilities. It also is an involved setup

- ☒ Export the data directly from DynamoDB to Amazon S3 with continuous backups. Turn on point-in-time recovery for the table.

This solution leverages DynamoDB's native continuous backup and export features, requiring minimal coding and having no impact on application availability or RCUs. Point-in-time recovery allows restoration to any point in the preceding 35 days.

- ☐ Configure Amazon DynamoDB Streams. Create an AWS Lambda function to consume the stream and export the data to an Amazon S3 bucket.

This solution requires coding and maintains logs only for 24 hours. The continuous backup process uses DynamoDB stream internally, this is not minimal coding.

- ☐ Create an AWS Lambda function to export the data from the database tables to Amazon S3 on a regular basis. Turn on point-in-time recovery for the table.

This solution necessitates coding for data export and regular scheduling. DynamoDB's managed solution provides better performance, no coding effort, and availability.

Overall Explanation:

The best solution is to utilize DynamoDB's built-in continuous backup and point-in-time recovery (PITR) features to export data to Amazon S3. This approach satisfies the requirements with minimal coding effort and ensures zero impact on application availability or RCUs. DynamoDB table export to S3 is a fully managed solution for exporting DynamoDB data to an Amazon S3 bucket at scale.

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